Factorised Cartesian Atomic Cluster Expansion for Low-Cost Machine-Learning Interatomic Potentials

Albert Linda*

Department of Materials Science and Engineering, Indian Institute of Technology Kanpur, Kalyanpur, Kanpur 208016, Uttar Pradesh, India

E-mail: albert.hzbn@gmail.com

Abstract

Machine-learning interatomic potentials can reach the accuracy of density functional theory at a fraction of its cost, but the most accurate equivariant models remain expensive, mainly because of the tensor operations they perform for every pair of neighbouring atoms. We present FACE, a factorised Cartesian atomic cluster expansion that removes this bottleneck. Each neighbour contributes only a small set of radial functions and messages, and all mixing of feature channels is done once per atom, so the cost per neighbour no longer grows with the size of the model. Custom GPU kernels evaluate the remaining steps, and long-range electrostatics, dispersion and short-range repulsion can be added through a single interface. We further show that training instabilities in this class of models originate in how the network weights are scaled, and we introduce a distillation scheme in which an ensemble of models generates additional, uncertainty-checked training data. On the rMD17 benchmark, with identical training budgets and hardware, FACE matches the accuracy of a 128-channel MACE model on aspirin at an 11 times lower inference cost; compared with a 64-channel MACE it is 5.7 times

faster, needs about four times less GPU memory and is more accurate on four of five molecules. Training costs about the same per step. A distilled single model reaches energy and force errors of 2.26 meV and 6.74 meV/Å on aspirin, close to the published MACE values (2.2 and 6.6), and runs stably in molecular dynamics.

1 Introduction

Machine-learning interatomic potentials (MLIPs) predict the energy and forces of an atomic configuration. Their accuracy approaches that of the electronic-structure method used to generate the training data, typically density functional theory (DFT), while their cost is orders of magnitude lower.[1,2] They now underpin molecular-dynamics studies that are out of reach for first-principles methods, and several reviews describe the state of the field.[3,4] Two developments define current practice. The first is to describe the environment of each atom by a body-ordered expansion of its neighbour density, that is, by a sum of terms that each depend on the positions of a fixed number of atoms (two-body, three-body and so on). Atom-centred symmetry functions,[5] the smooth overlap of atomic positions (SOAP),[6] the spectral neighbour analysis potential (SNAP),[7] moment tensor potentials (MTP)[8] and the atomic cluster expansion (ACE)[9–11] can all be read as members or truncations of one such expansion. ACE makes the construction systematic: a complete, body-ordered polynomial basis whose cost per atom is linear in the number of neighbours. The second development is message passing, in which atoms exchange learned features with their neighbours over several steps, so that each atom gradually sees a larger region. Invariant networks such as SchNet,[12] PhysNet[13] and DimeNet[14] were followed by equivariant networks, whose internal features rotate together with the structure: tensor field networks,[15] PaiNN,[16] NequIP,[17] Allegro,[18] equivariant transformers[19–21] and models with explicit many-body messages such as QuinNet[22] and GotenNet.[23] MACE[24] and graph ACE[25] unify the two developments: they are ACE expansions whose density is built from learned neighbour features, so that message passing increases the body order and the receptive field together.[26]

On the revised MD17 (rMD17) benchmark,[27,28] the leading equivariant models now differ by only a few meV/Å in force error. The practical bottlenecks have therefore shifted from accuracy on small molecules to cost, robustness and physics beyond locality. Four limitations motivate this work.

(i) Edge cost. An atomic structure can be viewed as a graph whose edges connect neighbouring atoms. The dominant cost of equivariant models is the tensor product computed on every edge, between the features of the neighbour and the spherical harmonics of the bond direction. Its cost scales with the maximal angular momentum and with the number of feature channels. SO(2) and Gaunt reductions[29,30] lower the angular cost, and Cartesian formulations (MTP, TensorNet,[31] CACE,[32] irreducible Cartesian tensor products[33]) remove Clebsch–Gordan coefficients altogether. All of them, however, keep channel-wide tensors on every edge, so that widening the model raises the dominant edge term. Systematic cost–accuracy comparisons[34] show that this trade-off, rather than the attainable accuracy, often decides which potential is used in practice.

(ii) Long-range physics. Local models cannot represent the total charge of a system or non-local charge transfer. Long-distance equivariant features,[35] charge-equilibration networks such as fourth-generation high-dimensional neural-network potentials,[36] models with explicit electronic degrees of freedom,[37] Ewald message passing[38] and latent Ewald summation[39] address this. The additional cost, and how such terms combine with a fast local model, are rarely quantified.

(iii) Training robustness. High-order polynomial features combined with normalisation layers can diverge. In our experience and in published training recipes, this is usually handled by tuning learning rates and schedules rather than diagnosed.

(iv) Data efficiency and uncertainty. With about one thousand reference structures, single models overfit. Deep ensembles[40] reduce the error and provide an uncertainty estimate but multiply the inference cost. Knowledge distillation[41] has been used to transfer the accuracy of large MLIPs to small ones;[42,43] how the teacher's sampling should be restricted

to the region where the teacher is reliable has received less attention.

In this work we present FACE (Factorised Cartesian Atomic Cluster Expansion). Its central idea is simple. In current equivariant models, every edge carries a tensor as wide as the model itself, so a more expressive model is proportionally more expensive on every edge. In FACE, each edge carries only a few radial functions and a short message; all widening happens once per atom, where it is cheap. We emphasise at the outset that FACE belongs to the ACE family. Its features are body-ordered contractions of neighbour-density moments, closely related to MTP and to MACE's symmetric contractions, and Section 3.2.3 makes this relation explicit. The contributions are in how this expansion is parameterised, implemented and trained:

1. *A cheaper edge.* A factorised edge density whose cost per edge does not depend on the model width, and a variant based on a canonical-polyadic (CP) decomposition that restores separate radial functions for every channel.

2. *Cheap tensor messages and fused kernels.* Equivariant messages between message-passing steps are computed by a single matrix multiplication followed by a fixed linear map, and custom GPU kernels evaluate the pooling over neighbours and all per-atom algebra.

3. *Pluggable physics.* One interface through which further energy terms are added, with their cost measured: long-range electrostatics based on latent charges, with charge equilibration and smooth particle-mesh Ewald summation (SPME);[44] dispersion; and the short-range repulsion of Ziegler, Biersack and Littmark (ZBL).

4. *Stable training.* A diagnosis of training divergence as a problem of how the weights are scaled, and a fix (unit-variance parameterisation).

5. *Better single models from ensembles.* A distillation scheme in which an ensemble of models (the teacher) runs molecular dynamics, labels the visited structures on which

its members agree, and trains a single model (the student), which is finally fine-tuned on the DFT data alone; students join the next teacher ("born-again" iteration[45]).

All comparisons with MACE that support efficiency claims use models trained on identical splits and epoch budgets and timed on the same GPU; comparisons with published numbers are labelled as such.

The paper is organised as follows. Section 2 summarises the atomic cluster expansion and its message-passing form, and identifies where the cost arises. Section 3 derives the FACE model, its relation to ACE, the physics operators and a cost model, and Section 4 describes the training procedure and the distillation protocol. Section 5 describes the implementation, its extension points and the verification suite. Section 6 gives the computational details, and Section 7 reports accuracy (including ablations, uncertainty and distillation), computational cost, training stability, long-range benchmarks and molecular-dynamics tests. Section 8 discusses the position of FACE among related models and its limitations.

2 Background: atomic cluster expansion and message passing

This section summarises the two ideas on which FACE builds and shows where their computational cost arises. Readers familiar with ACE and MACE can continue with Section 3.

2.1 Atomic cluster expansion

In ACE[9,10] the energy is a sum of atomic energies, $E = \sum_i E_i$, and each E_i is expanded in body-ordered features of the neighbour density of atom i. The one-particle basis is projected onto the neighbourhood,

$$A_{i,znlm} = \sum_{j \in \mathcal{N}(i)} \delta_{z,Z_j} \, R_{nl}^{Z_i Z_j}(r_{ij}) \, Y_{lm}(\hat{\boldsymbol{r}}_{ij}), \tag{1}$$

where $\mathcal{N}(i)$ are the neighbours within the cutoff r_c, R_{nl} are radial functions and Y_{lm} spherical harmonics. Products of ν such projections, $\boldsymbol{A}_{i,\boldsymbol{v}} = \prod_{t=1}^{\nu} A_{i,v_t}$, are $(\nu+1)$-body functions of the environment, because each projection is a single sum over neighbours. Rotation-invariant features $B_{i,\boldsymbol{v}}$ follow by coupling the angular indices with generalised Clebsch–Gordan coefficients, and $E_i = \sum_{\boldsymbol{v}} c_{\boldsymbol{v}} B_{i,\boldsymbol{v}}$ or a nonlinear function of the B. The essential computational property is that the expensive many-body structure is formed per atom from the projections A, which cost only one sum over neighbours. MTP[8] uses the same construction with Cartesian moment tensors $\sum_j f_\mu(r_{ij})\, \hat{\boldsymbol{r}}_{ij}^{\otimes\nu}$ in place of $R_{nl}Y_{lm}$.

2.2 Message passing as a learned density

MACE[24] replaces the fixed chemical index z in Equation 1 by learned neighbour features $h_{j,k}$ with $k = 1, \ldots, C$ channels. In the first layer,

$$A^{(1)}_{i,klm} = \sum_j R^{(1)}_{kl}(r_{ij})\, Y_{lm}(\hat{\boldsymbol{r}}_{ij}) \sum_{\tilde{k}} W_{k\tilde{k}} h^{(0)}_{j,\tilde{k}}, \tag{2}$$

and in later layers the neighbour features h_j are themselves equivariant, so that the edge operation becomes a channel-wise tensor product between h_j and $Y(\hat{\boldsymbol{r}}_{ij})$ weighted by learned radial functions. A symmetric contraction then forms body-ordered products per atom, as in ACE. The resulting cost has two parts. The per-atom contraction scales with C and with the correlation order. The per-edge tensor product scales with C multiplied by the number of coupling paths between the irreducible representations of h_j, Y and the output. For large systems the number of edges exceeds the number of atoms by the mean neighbour number $\bar{M}$, typically 10–40, so the per-edge term dominates.

2.3 Cartesian and spherical tensors

A symmetric Cartesian tensor of rank l has $(l+1)(l+2)/2$ independent components and decomposes into irreducible parts of degree $l, l-2, \ldots$: $\mathrm{Sym}^2(\mathbb{R}^3) = \mathbf{0} \oplus \mathbf{2}$ and $\mathrm{Sym}^3(\mathbb{R}^3) =$

$\mathbf{1} \oplus \mathbf{3}$. The symmetric powers $\hat{\boldsymbol{u}}^{\otimes l}$ of a unit vector for $l = 0, \ldots, 3$ therefore span the same function space as the spherical harmonics with $l \leq 3$, plus redundant trace components that duplicate lower degrees. Cartesian models avoid Clebsch–Gordan coefficients because contractions of Cartesian tensors are invariant whenever all indices are paired by Kronecker deltas. TensorNet, CACE and ICTP use this property; MTP is its original form.

3 The FACE model

3.1 Overview and notation

FACE computes the total energy as

$$E = \sum_i \left[E^0_{Z_i} + s \sum_{t=1}^{T} \varepsilon_i^{(t)} \right] + \sum_k \sum_i E_{k,i}, \tag{3}$$

where E^0_Z are element reference energies, s is a fixed output scale, $\varepsilon_i^{(t)}$ is the atomic-energy contribution of message-passing step t (we use $T = 2$), and $E_{k,i}$ are per-atom energies of the physics operators of Section 3.6. Forces are $\boldsymbol{F}_i = -\partial E / \partial \boldsymbol{r}_i$ by automatic differentiation, so the model is conservative by construction. Each atom carries scalar features $\boldsymbol{h}_i^{(t)} \in \mathbb{R}^C$, initialised as a learned embedding of the element, $\boldsymbol{h}_i^{(0)} = \boldsymbol{e}_{Z_i}$. Step t performs four operations: (1) it builds Cartesian moments $A_i^{(l,t)}$, $l = 0, \ldots, 3$, from the neighbours (Sections 3.2.1–3.3.2); (2) it forms body-ordered invariants from the moments of all steps so far (Section 3.4); (3) it updates the scalar features and emits $\varepsilon_i^{(t)}$; and (4) for $t < T$ it forms equivariant vectors and matrices that are sent to the neighbours in the next step. Figure 1 summarises the data flow, and Table 1 collects the symbols used below.

In plain terms, a single step works as follows. For every neighbour, the model computes a handful of numbers that depend on the distance (radial functions) and a short summary of what the neighbour is (a message). These are multiplied by simple polynomials of the bond direction and summed over all neighbours, giving a compact description of the atom's

surroundings. Everything that is expensive, such as mixing many feature channels and multiplying descriptions together to capture three-, four- and five-body correlations, is then done once per atom rather than once per neighbour. In the second step, atoms additionally send each other small vectors and matrices that describe the orientation of their own environments.

Table 1: Main symbols and their values in the flagship configuration.

symbol	meaning	flagship
i, j	central atom, neighbour	
r_{ij}, $\hat{\boldsymbol{u}}_{ij}$	distance and unit vector from i to j	
r_c	cutoff radius	4.5 Å
$\bar{n}$	mean number of neighbours in the training set	12.3 (aspirin)
Y_x	the 20 direction monomials of ranks $l = 0$–3, Equation 4	
B	number of learned radial functions ϕ_b per edge	16
K	length of the neighbour message m_{jk}	32
C	node width (feature channels per atom)	96
K_d	channels of the vectors $\boldsymbol{V}$ and matrices $\boldsymbol{M}$ sent between steps	16
T	number of message-passing steps	2
$A_i^{(l)}$	Cartesian moments (environment descriptors) of rank l	
P, Q, S	learned projections of the moments used in products	
$\boldsymbol{h}_i$	scalar features of atom i	
$\varepsilon_i^{(t)}$	atomic energy contribution of step t	

3.2 Local environment descriptors

3.2.1 Cartesian moments in symmetric storage

For atom i with neighbours j within the cutoff r_c, let $\boldsymbol{r}_{ij} = \boldsymbol{r}_j - \boldsymbol{r}_i$, $r_{ij} = |\boldsymbol{r}_{ij}|$ and $\hat{\boldsymbol{u}}_{ij} = \boldsymbol{r}_{ij}/r_{ij}$. Each direction is lifted to its symmetric outer powers $\hat{\boldsymbol{u}}^{\otimes l}$, $l = 0, \ldots, 3$, stored by their $1 + 3 + 6 + 10 = 20$ independent components

$$Y_x(\hat{\boldsymbol{u}}) \in \{1;\ u_a;\ u_a u_b\ (a \le b);\ u_a u_b u_c\ (a \le b \le c)\}, \qquad x = 1, \ldots, 20. \tag{4}$$

Under a rotation or reflection $Q \in O(3)$, $\hat{\boldsymbol{u}}^{\otimes l} \mapsto Q^{\otimes l}\hat{\boldsymbol{u}}^{\otimes l}$, so any linear map that acts only on channel indices commutes with Q and preserves equivariance, and every full index contraction

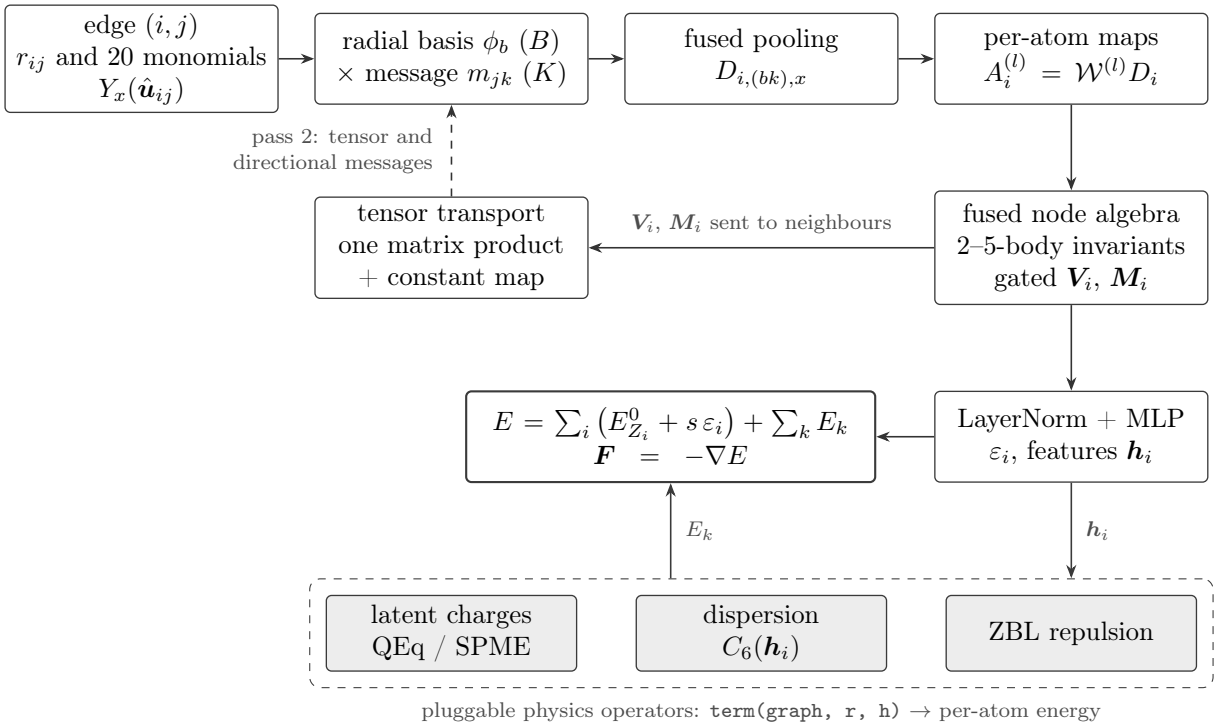


Figure 1: Structure of FACE. White boxes: local ACE-family model (one pass shown; the second pass adds directional and tensor messages, dashed arrow). Grey boxes: pluggable physics operators, which receive the final features $\boldsymbol{h}_i$ and add per-atom energies. The edge cost depends on B and K but not on the node width C.

of products is invariant. Compared with the 16 spherical harmonics with $l \leq 3$, the 20 monomials contain four redundant trace components (the trace of the rank-2 tensor and the trace vector of the rank-3 tensor). This redundancy is harmless for learning and avoids projections onto irreducible parts. Full tensors are formed only per atom, where contractions are evaluated. Contractions in symmetric storage weight each component by its multiplicity w_x (1,1,1,2,2,2 for rank 2 and 1,1,1,3,...,3,6 for rank 3), e.g. $P^{(2)}{:}Q^{(2)} = \sum_x w_x P_x^{(2)} Q_x^{(2)}$. In words, the monomials Y_x encode the direction of one bond up to third order; summed over neighbours with suitable weights, they describe the angular shape of an environment, much as a multipole expansion describes the shape of a charge distribution.

3.2.2 Factorised edge densities

Each edge carries two small objects. The first is a learned radial basis,

$$\phi_b(r_{ij}; Z_i, Z_j) = \mathrm{MLP}\big[\,\mathrm{SiLU}\big(W\,\mathbf{g}(r_{ij}) f_c(r_{ij}) + \boldsymbol{e}_{Z_i Z_j}\big)\big]_b\, f_c(r_{ij}), \qquad b = 1, \ldots, B, \tag{5}$$

with a multilayer perceptron (MLP), eight Bessel functions $\mathbf{g}$, the polynomial envelope f_c of degree 6, an element-pair embedding $\boldsymbol{e}_{Z_i Z_j}$ and SiLU activations.[46] The second is a low-rank neighbour message $m_{jk} = (W_m \boldsymbol{h}_j)_k$, $k = 1, \ldots, K$, computed from the scalar features of the neighbour. The atomic density

$$D_{i,(bk),x} = \frac{1}{\bar{n}} \sum_j \phi_b(r_{ij})\, m_{jk}\, Y_x(\hat{\boldsymbol{u}}_{ij}) \tag{6}$$

is a single batched matrix product of size $(BK) \times 20$ per atom, where $\bar{n}$ is the mean number of neighbours in the training set. The rank-l moments with C channels are obtained by channel maps that act per atom,

$$A_{i,c,x}^{(l)} = s_{Z_i,l,c} \sum_{b,k} \mathcal{W}_{c,(bk)}^{(l)}\, D_{i,(bk),x}, \qquad x \in \text{rank } l, \tag{7}$$

with a learned centre-element scale $s_{Z_i,l,c}$. The per-edge work is therefore $O(BK \cdot 20)$ and independent of C. In words, instead of giving every feature channel its own radial function on every edge, all channels share B radial functions and a K-component message; the per-atom map $\mathcal{W}^{(l)}$ then combines the pooled products into as many channels as desired.

3.2.3 Relation to the ACE density

Substituting Equation 6 into Equation 7 gives

$$A^{(l)}_{i,c,x} = \frac{s_{Z_i,l,c}}{\bar{n}} \sum_j \mathcal{R}^{(l)}_c(r_{ij}; Z_i, Z_j, \boldsymbol{h}_j)\, Y_x(\hat{\boldsymbol{u}}_{ij}), \qquad \mathcal{R}^{(l)}_c = \sum_{b=1}^{B} \sum_{k=1}^{K} \mathcal{W}^{(l)}_{c,(bk)}\, \phi_b(r_{ij}; Z_i, Z_j)\, m_{jk}. \tag{8}$$

This is an ACE (or MACE) density, Equation 1, with a channel- and rank-dependent radial–chemical function $\mathcal{R}^{(l)}_c$. In the first step $\boldsymbol{h}_j = \boldsymbol{e}_{Z_j}$, so $\mathcal{R}^{(l)}_c$ depends only on (r_{ij}, Z_i, Z_j) and Equation 8 is an ACE projection with a learned chemical embedding, as in tensor-reduced ACE.[47] The factorisation restricts the C functions $\mathcal{R}^{(l)}_c$ to a common BK-dimensional space spanned by products of B radial functions and K message components. This is the source of the efficiency: the edge evaluates and pools the BK basis products once, and the per-atom map $\mathcal{W}^{(l)}$ produces any number of channels from them. It is also the main restriction: if the optimal $\mathcal{R}^{(l)}_c$ need more than BK independent combinations, accuracy is lost (Section 7.1.2).

3.2.4 CP variant

In FACE-CP the map $\mathcal{W}^{(l)}$ is replaced by a canonical-polyadic (CP) decomposition[48] of rank C, which is equivalent to per-(rank, channel) radial functions $R_{l,c}(r)$ multiplying a channel-wise message μ_{jc},

$$A^{(l)}_{i,c,x} = \sum_{c'} \mathcal{M}^{(l)}_{cc'} \frac{s_{Z_i,l,c'}}{\bar{n}} \sum_j R_{l,c'}(r_{ij})\, \mu_{jc'}\, Y_x(\hat{\boldsymbol{u}}_{ij}), \tag{9}$$

followed by a per-atom channel mixing $\mathcal{M}^{(l)}$. In words, the CP variant gives every channel its own radial function again, which is more flexible but costs more per edge. This is the form of MACE's first-layer density. The edge cost is now $4C$ per edge and grows with C,

but it is evaluated as one product of the 20 monomials with a contiguous $(4C)$-wide edge tensor, keeping the rank-diagonal blocks of the small output.

3.3 Messages between steps

3.3.1 Directional messages

From the second step on, each atom also carries K_d channels of equivariant vectors $\boldsymbol{V}_i$ and symmetric matrices $\boldsymbol{M}_i$ (Section 3.5). The scalar message is modulated by bond-projected invariants of both end points,

$$m_{jk} \leftarrow m_{jk} + \left[W_d\, \boldsymbol{d}_{ij}\right]_k, \qquad \boldsymbol{d}_{ij} = \left(\hat{\boldsymbol{u}}_{ij} \cdot \boldsymbol{V}_j,\ \hat{\boldsymbol{u}}_{ij}^\top \boldsymbol{M}_j \hat{\boldsymbol{u}}_{ij},\ \hat{\boldsymbol{u}}_{ij} \cdot \boldsymbol{V}_i,\ \hat{\boldsymbol{u}}_{ij}^\top \boldsymbol{M}_i \hat{\boldsymbol{u}}_{ij}\right), \tag{10}$$

with one entry per channel $k \le K_d$, so $\boldsymbol{d}_{ij} \in \mathbb{R}^{4K_d}$. The angle between the bond and each atom's own environment thus enters the second-step density without additional tensor products. All four entries are invariant, so Equation 6 remains equivariant.

3.3.2 Tensor transport by a constant linear map

Neighbours also send their equivariant tensors directly. The Cartesian analogues of the $1 \otimes 0 \to 1$, $2 \otimes 1 \to 1$, $2 \otimes 0 \to 2$ and $1 \otimes 1 \to 2$ couplings of spherical models are

$$\sum_j \phi_b \boldsymbol{V}_j, \qquad \sum_j \phi_b\, \boldsymbol{M}_j \hat{\boldsymbol{u}}_{ij}, \qquad \sum_j \phi_b \boldsymbol{M}_j, \qquad \sum_j \phi_b\, \mathrm{sym}(\hat{\boldsymbol{u}}_{ij} \otimes \boldsymbol{V}_j), \tag{11}$$

with $\mathrm{sym}(\boldsymbol{a} \otimes \boldsymbol{b})_{ab} = \frac{1}{2}(a_a b_b + a_b b_a)$. Optional rank-3 couplings add $\sum_j \phi_b\, \mathrm{sym}(\hat{\boldsymbol{u}} \otimes \boldsymbol{M}_j)$ and $\sum_j \phi_b\, \mathrm{sym}(\hat{\boldsymbol{u}} \otimes \hat{\boldsymbol{u}} \otimes \boldsymbol{V}_j)$. A naive implementation forms each product on every edge. Instead, note that every term in Equation 11 is linear in the neighbour tensor $T_j = [\boldsymbol{V}_j, \boldsymbol{M}_j]$ (9 components per channel) and polynomial of degree at most one in $\hat{\boldsymbol{u}}$; for example $(\boldsymbol{M}_j \hat{\boldsymbol{u}})_a = \sum_b M_{j,ab} u_b$ and $\mathrm{sym}(\hat{\boldsymbol{u}} \otimes \boldsymbol{V}_j)_{ab} = \frac{1}{2}(u_a V_{j,b} + u_b V_{j,a})$. All of them are therefore fixed linear

combinations of the sums

$$S_{i,(p,b),(k,y)} = \frac{1}{\bar{n}} \sum_j Y_p(\hat{\boldsymbol{u}}_{ij})\, \phi_b(r_{ij})\, T_{j,k,y}, \qquad p \in \{1, u_x, u_y, u_z\}, \quad y = 1, \ldots, 9, \tag{12}$$

which is one batched matrix product of size $(4B) \times (9K_d)$ per atom. Every transport moment is then obtained by a constant (36×18) matrix acting on the index pair (p, y); its 18 outputs are the 3+3 vector and 6+6 matrix components of Equation 11. With the rank-3 option, p runs over the 10 monomials of degree ≤ 2 and the constant map is (90×38), adding two rank-3 outputs of 10 components each. The results are mixed into the C moment channels of the matching rank by per-atom maps from the $B \cdot 2K_d$ (basis, channel) pairs. No per-edge intermediate tensor is formed, and the map is verified against explicit per-edge evaluation to 10^{-16} (Section 5.4). In words, instead of forming several small tensor products on every edge, the model sums each neighbour's vector and matrix weighted by distance and by the bond direction, and reads off all required combinations afterwards from a fixed table of coefficients.

3.4 Body-ordered invariants and readout

The moments describe the environment but still rotate with it. To obtain an energy, which must not change when the structure is rotated, they are multiplied together and all Cartesian indices are summed out. Each resulting number measures, for example, how strongly two directional features of the environment are aligned. In step t, the moments of all steps $1, \ldots, t$ are concatenated along the channel index. Per rank l, three learned linear projections produce $P^{(l)}_{i,c} = \sum_{c'} W^{P,l}_{cc'} A^{(l)}_{i,c'}$ and similarly $Q^{(l)}$ and $S^{(l)}$, each with C channels. Optionally, the projections are multiplied by centre-element-dependent gates $g_{Z_i,l,c}$, which gives element-specific product weights as in MACE. The projections are contracted channel-diagonally into

nine invariant families (indices a, b, d, e are Cartesian; channel index omitted):

$$
\begin{aligned}
&\text{2-body:} \quad I_1 = P^{(0)}, \\
&\text{3-body:} \quad I_2 = P^{(1)}_a Q^{(1)}_a, \quad I_3 = P^{(2)}_{ab} Q^{(2)}_{ab}, \quad I_4 = P^{(3)}_{abd} Q^{(3)}_{abd}, \\
&\text{4-body:} \quad I_5 = P^{(1)}_a Q^{(2)}_{ab} S^{(1)}_b, \quad I_6 = P^{(2)}_{ab} Q^{(2)}_{bd} S^{(2)}_{da}, \quad I_7 = P^{(3)}_{abd} Q^{(2)}_{ab} S^{(1)}_d, \quad I_8 = P^{(3)}_{abd} Q^{(3)}_{abe} S^{(2)}_{de}, \\
&\text{5-body:} \quad I_9 = P^{(3)}_{abd} Q^{(1)}_a S^{(1)}_b Q^{(1)}_d,
\end{aligned}
\tag{13}
$$

with summation over repeated indices. Each family is a full contraction with Kronecker deltas and hence invariant under rotations. Under reflection, a rank-l moment acquires the sign $(-1)^l$; the total rank of every family is even (0, 2, 4, 6, 4, 6, 6, 8, 6), so all invariants are also parity-even and the energy is $O(3)$-invariant. In the first step, each moment is a single sum over neighbours, so a product of ν moments is a $(\nu + 1)$-body function of the environment; this gives the body orders listed. The families are a subset of the ACE product basis up to correlation order four with $l \leq 3$. They are not complete, and we make no completeness claim (see also reference 49); the learned projections P, Q, S supply the cross-channel couplings that a channel-diagonal contraction alone would miss.

The $9C$ invariants are normalised by layer normalisation[50] and passed, together with the current scalar features, to an MLP of widths $(9C + C) \to 2C \to 2C \to C + 1$ (four further optional inputs are used only with moment normalisation and are zero here). The first output is $\varepsilon_i^{(t)}$ and the remaining C outputs update the features residually, $\boldsymbol{h}_i^{(t)} = \boldsymbol{h}_i^{(t-1)} + \Delta\boldsymbol{h}_i$.

3.5 Gated equivariant products

For the next step, each atom forms equivariant vectors and symmetric matrices from the same projections,

$$
\begin{aligned}
\boldsymbol{V} &= a_0 P^{(1)} + a_1\, Q^{(2)} S^{(1)} + a_2\, P^{(3)} : Q^{(2)} + a_3\, P^{(2)} Q^{(1)}, \\
\boldsymbol{M} &= \mathrm{sym}\big(b_0 P^{(2)} + b_1\, P^{(1)} \otimes Q^{(1)} + b_2\, P^{(2)} Q^{(2)} + b_3\, P^{(3)} \cdot S^{(1)}\big),
\end{aligned}
\tag{14}
$$

with per-channel path weights a_n, b_n (initialised to one), and multiplies them channel-wise by gates that are linear functions of the invariants, $\boldsymbol{V} \leftarrow \boldsymbol{g}^V \odot \boldsymbol{V}$, $\boldsymbol{g}^V = W_g I + \mathbf{1}$ (and likewise for $\boldsymbol{M}$; W_g is initialised to zero). The gated tensors are polynomials of degree up to three in the moments. They are projected to K_d channels for the directional messages and the transport of the next step, so the body order of the second-step features grows multiplicatively with that of the first step, as in MACE. Without these products, the transported tensors would be linear in the first-step moments and the second step would add body order only through the scalar messages.

3.6 Physics operators

Every operator E_k in Equation 3 has the same interface: it receives the graph, the interatomic distances and the final scalar features $\boldsymbol{h}_i$, and returns per-atom energies. Forces follow by automatic differentiation through the operator, including any linear solves.

3.6.1 Latent-charge electrostatics

A small MLP maps $\boldsymbol{h}_i$ to an electronegativity χ_i and a hardness $J_i = \mathrm{softplus}(\tilde{J}_i) + 0.1\ \mathrm{eV}/e^2$. Atoms carry Gaussian charges of width $\sigma = 0.5$ Å. Two such charges interact through the smeared kernel $\mathrm{erf}(r/\gamma)/r$ with $\gamma = 2\sigma$, which defines the Coulomb matrix A_{ij} ($A_{ii} = 0$). The electrostatic energy is

$$E_{\mathrm{es}} = \sum_i \chi_i q_i + \tfrac{1}{2} \sum_i J_i q_i^2 + \tfrac{1}{2} k_e \sum_{ij} q_i A_{ij} q_j, \qquad \sum_i q_i = Q, \tag{15}$$

with $k_e = 14.3996\ \mathrm{eV\,Å}/e^2$ and the total charge Q of the structure as an input. In the *coulomb* mode the charges follow from a global chemical potential without Coulomb coupling,

$$q_i = -\frac{\chi_i - \mu}{J_i}, \qquad \mu = \frac{Q + \sum_i \chi_i / J_i}{\sum_i 1/J_i}, \tag{16}$$

which costs $O(N)$ and redistributes charge over the whole structure through μ. In the *qeq* mode the charges minimise Equation 15 including the Coulomb term (charge equilibration[36,51]), which requires the linear system

$$\begin{pmatrix} k_e A + \operatorname{diag}(\boldsymbol{J}) & \mathbf{1} \\ \mathbf{1}^{\top} & 0 \end{pmatrix} \begin{pmatrix} \boldsymbol{q} \\ -\mu \end{pmatrix} = \begin{pmatrix} -\boldsymbol{\chi} \\ Q \end{pmatrix}. \tag{17}$$

For periodic systems, $\frac{1}{2}\boldsymbol{q}^{\top} A \boldsymbol{q}$ is evaluated by Ewald summation[52] with a second splitting width γ',

$$\tfrac{1}{2}\boldsymbol{q}^{\top} A \boldsymbol{q} = \tfrac{1}{2} \sum_{i \neq j, \boldsymbol{n}} q_i q_j \frac{\operatorname{erf}(r/\gamma) - \operatorname{erf}(r/\gamma')}{r} + \frac{2\pi}{V} \sum_{\boldsymbol{k} \neq 0} \frac{e^{-\gamma'^2 k^2/4}}{k^2} \left|S(\boldsymbol{k})\right|^2 - \frac{1}{\sqrt{\pi}\gamma'} \sum_i q_i^2 - \frac{\pi \gamma'^2}{2V} Q^2, \tag{18}$$

with $r = |\boldsymbol{r}_{ij} + \boldsymbol{n}|$ over lattice vectors $\boldsymbol{n}$, the structure factor $S(\boldsymbol{k}) = \sum_i q_i e^{i\boldsymbol{k}\cdot\boldsymbol{r}_i}$ and a neutralising-background term for charged cells. We use $\gamma' = \max(r_c/3.5,\, 1.05\gamma)$, so that the real-space sum converges within the model cutoff and reuses its neighbour list. The reciprocal sum is evaluated by SPME:[44,53] charges are spread on a grid of spacing $\gamma'/4$ with order-4 cardinal B-splines, the grid is Fourier transformed, multiplied by the Ewald kernel and the Euler-spline moduli, and the energy is accumulated, at a cost of $O(N + G \log G)$ for G grid points.

3.6.2 Dispersion

The dispersion operator is

$$E_{\mathrm{disp}} = -\frac{1}{2} \sum_{i \neq j} \frac{C_{6,ij}}{r_{ij}^6 + R_{0,ij}^6} S(r_{ij}), \qquad C_{6,ij} = \sqrt{C_{6,i} C_{6,j}}, \quad C_{6,i} = C_{6,Z_i}^{\mathrm{free}} \exp(\boldsymbol{w} \cdot \boldsymbol{h}_i), \tag{19}$$

with free-atom C_6 coefficients and van der Waals radii R_Z from Tkatchenko and Scheffler,[54] the geometric combination rule, rational Becke–Johnson-type damping[55] $R_{0,ij} = a_1(R_{Z_i} + R_{Z_j})/2 + a_2$ with learned positive a_1, a_2, and a smooth switch S that goes from one to zero

between 8 and 10 Å. The environment modulation $\exp(\boldsymbol{w} \cdot \boldsymbol{h}_i)$ starts at one ($\boldsymbol{w} = 0$) and plays the role of the coordination dependence of D3[56] or the Hirshfeld rescaling of TS. The sum uses a separate pair list with a 10 Å cutoff.

3.6.3 Short-range repulsion

The universal Ziegler–Biersack–Littmark (ZBL) potential,[57]

$$E_{\mathrm{ZBL}} = \frac{1}{2} \sum_{i \neq j} \frac{k_e Z_i Z_j}{r_{ij}} \, \varphi\!\left(\frac{r_{ij}}{a_{ij}}\right) S_{\mathrm{Z}}(r_{ij}), \qquad a_{ij} = \frac{0.4685 \text{ Å}}{Z_i^{0.23} + Z_j^{0.23}}, \tag{20}$$

with the standard four-exponential screening function φ, is switched off between 0.6 and 1.2 Å by $S_{\mathrm{Z}}(r) = 1 - t^3(10 - 15t + 6t^2)$, $t = (r - 0.6 \text{ Å})/0.6 \text{ Å}$, which is C^2-continuous. It only acts at distances that are not sampled in typical training data and prevents unphysical attraction there.

3.7 Cost model

Let N be the number of atoms and $\bar{M}$ the mean number of neighbours. Table 2 lists the multiply–add operations per edge and per atom of one second-step pass of the flagship ($C = 96$, $B = 16$, $K = 32$, $K_d = 16$). The per-edge work, about 2.7×10^4 operations, is independent of C; it consists of the radial network, the density product and the transport product. The per-atom work is proportional to C and consists of dense matrix products: the channel maps $\mathcal{W}^{(l)}$, the projections, the transport mixing and the readout MLP. Counted in operations, both parts are of similar size at the 36 neighbours per atom of our condensed timing boxes. They differ in hardware efficiency. The per-atom maps are large dense matrix products, which GPUs execute efficiently, whereas edge operations involve gathers and irregular memory access and are bound by memory bandwidth. Widening the model therefore raises only the efficient part of the cost: doubling C from 64 to 128 increased the measured time by 29% at 4.5k atoms, whereas the same change in MACE (MACE-small to MACE-medium) increased

it by 89% (Table 8); Section 7.2.4 measures this over a wider range of widths. Memory follows the same pattern; the widest per-edge tensors are BK (pooled inside the fused kernel without being stored) and $9K_d$.

Table 2: Multiply–add operations of one second-step pass of the flagship ($C = 96$, $B = 16$, $K = 32$, $K_d = 16$). The first step has no directional or transport terms.

operation	size	operations
per edge (independent of C)		
radial network, Equation 5	$8 \to 64 \to 64 \to B$	5.6×10^3
density product, Equation 6	$B \cdot K \cdot 20$	1.0×10^4
directional invariants, Equation 10	$2 \cdot 9K_d + 4K_d \cdot K$	2.3×10^3
transport product, Equation 12	$4B \cdot 9K_d$	9.2×10^3
total per edge		2.7×10^4
per atom ($\propto C$)		
channel maps, Equation 7	$BK \cdot C \cdot 20$	9.8×10^5
projections P, Q, S	$2C \cdot 3C \cdot 20$	1.1×10^6
transport mixing	$2K_dB \cdot C \cdot 9$	4.4×10^5
invariants and gated products		$\approx 3 \times 10^4$
readout MLP	$10C \to 2C \to 2C \to C$	2.4×10^5

4 Training and distillation

This section describes how the models are trained, why the weights are parameterised as they are, and the distillation protocol that turns an ensemble of models into a better single model.

4.1 Reference energies and output scaling

For single-molecule data sets the mean training energy is subtracted in double precision before training in single precision; for data with varying composition, element reference energies E_Z^0 are obtained by least squares on the training set. The output scale s in Equation 3 is the standard deviation of the training force components, and $\bar{n}$ is the mean neighbour number of the training set (12.3 for aspirin at $r_c = 4.5$ Å).

4.2 Loss and optimisation

The loss is

$$\mathcal{L} = w_E \Big\langle \Big(\frac{E - E^{\mathrm{ref}}}{N} \Big)^2 \Big\rangle + w_F \big\langle |\boldsymbol{F}_i - \boldsymbol{F}_i^{\mathrm{ref}}|^2 \big\rangle \tag{21}$$

with $(w_E, w_F) = (1, 1000)$ for the first 75% of epochs and $(1000, 100)$ afterwards. We use Adam[58] with the AMSGrad correction[59] and decoupled weight decay (set to zero),[60] a one-cycle learning-rate schedule[61] with peak 0.04, batch size 5, gradient-norm clipping at 10 and an exponential moving average (EMA, decay 0.99) of the parameters and of all buffers, which must be copied explicitly. Training with forces requires second derivatives, because the loss depends on $\partial E / \partial \boldsymbol{r}$; the model is therefore built from twice-differentiable operations.

4.3 Unit parameterisation

Every linear layer stores weights $W \sim \mathcal{N}(0, 1)$ and applies the factor $1/\sqrt{n_{\mathrm{in}}}$ in the forward pass, $\boldsymbol{y} = W\boldsymbol{x}/\sqrt{n_{\mathrm{in}}} + \boldsymbol{b}$, as in e3nn[62] and related parameterisation analyses.[63] The function at initialisation is the same as with the default initialisation, which stores W at scale $1/\sqrt{n_{\mathrm{in}}}$ directly. The difference lies in the optimiser dynamics. Adam normalises each update by a running estimate of the gradient magnitude, so that the step on a stored weight is of order the learning rate η, independent of the gradient scale. With the default parameterisation the relative change of an effective weight per step is therefore $\eta/(1/\sqrt{n_{\mathrm{in}}}) = \eta\sqrt{n_{\mathrm{in}}}$, which grows with the layer width: for the 1024-wide edge maps and $\eta = 0.01$ it is about 30%. With unit parameterisation the relative change is η for every layer. Wide layers in this model sit directly before high-order polynomial products, which amplify relative changes of the moments by up to the fifth power. Section 7.3 shows that this explains the divergences we observed.

Recovery from non-finite steps. After 20 consecutive steps with a non-finite loss or gradient, the weights are restored from the EMA, the optimiser state is reset and the learning

rate is halved.

4.4 Uncertainty-gated ensemble distillation

An ensemble of M_{T} independently trained models serves as teacher. For a configuration $\boldsymbol{x}$, the teacher prediction is the member mean, and its uncertainty is the force spread

$$\sigma_F(\boldsymbol{x}) = \frac{1}{3N} \sum_{i,a} \mathrm{std}_m \big[F_{ia}^{(m)}(\boldsymbol{x}) \big], \tag{22}$$

the standard deviation over members averaged over atoms and Cartesian components. The protocol is:

1. *Teacher dynamics.* All training structures are propagated simultaneously by Langevin dynamics (BAOAB splitting scheme,[64] time step 0.5 fs, 400 steps, $T = 500$ K) with the ensemble-mean forces.

2. *Labelling and gating.* Every 20 steps, snapshots are labelled with the ensemble-mean energy and forces. Snapshots with σ_F larger than three times the mean spread on held-out data are discarded.

3. *Student training.* A student is trained on the DFT data, repeated four times per epoch, together with the teacher data, using the same number of optimiser steps as a 400-epoch run on DFT data alone.

4. *DFT-only fine-tune.* The student is fine-tuned on DFT data only (60 epochs, peak learning rate 4×10^{-3}), and optionally once more with an energy-weighted loss ($w_E = 1000$, $w_F = 10$, peak learning rate 2×10^{-3}).

5. *Iteration.* Students join the next teacher ("born-again" iteration[45]); existing teacher-MD configurations are relabelled by the new teacher.

No additional DFT calculations are performed, and test structures are never used. The gate in step 2 matters because the teacher's accuracy is only known on configurations similar to the training data; the threshold of three times the test-set spread keeps the teacher data within this region (Section 7.1.5). The final DFT-only fine-tune removes any systematic bias that the teacher labels introduce. Distillation of MLIPs is established;[41–43] the elements we add are the thermal teacher-MD sampling with an uncertainty gate, iterated teachers, and the final DFT-only and energy-weighted fine-tunes.

5 Implementation

5.1 Software design and extension points

FACE is implemented in PyTorch,[65] with CUDA kernels compiled at run time through CuPy[66] and NVIDIA's run-time compiler (NVRTC), so that no host C++ compiler is needed. The model is one module whose components are selected by constructor arguments and switches (Table 3); all switches are covered by the test suite. Configurations differ only in these settings, so the flagship, the CP variant and the ablations of Section 7.1.2 are produced by the same code.

Physics operators are `torch.nn.Module` objects with the method `forward(graph, r, h)` that returns a per-atom energy (Figure 1). Adding an operator requires implementing this one method and registering its name; the dispersion operator, for example, is 48 lines including its parameter tables. Because forces come from automatic differentiation of the total energy, no force code is written for new operators. Operators that need pairs beyond the model cutoff, such as dispersion, request a separate long-range pair list from the data pipeline. Charges are written back to the graph, so later operators or analyses can use them.

For simulations, an ASE[67] calculator wraps a trained model, and an ensemble calculator returns the member mean together with per-atom force spreads. Trained checkpoints store the model configuration, the reference energies and the normalisation constants, so a model

is restored from one file.

Table 3: Model options. All combinations used in this work are produced by the same code.

option	effect	default (flagship)
`factor`, B, K	factorised edge density, Equations 6–7	on, 16, 32
`cp`	CP density, Equation 9	off
`transport`	tensor transport, Equation 11	on
`transport3`	rank-3 transport couplings	off
`eqmsg`	gated equivariant products, Equation 14	on
`elem_proj`	element-dependent product gates	off
`max_body`	truncate invariant families by body order	5
`poly_readout`	linear readout in all but the last step	off
`terms`	physics operators: `coulomb`, `qeq`, `disp`, `zbl`	none
`fuse`, `compile_radial`	fused kernels and spline radial tables (inference)	on for MD

5.2 Neighbour lists and batching

Graphs are stored as a padded neighbour matrix of shape $N \times M_{\max}$ with a mask and lattice shift vectors, built with the ASE neighbour list. This layout makes every per-edge operation a dense tensor operation and turns the pooling of Equations 6 and 12 into batched matrix products over the neighbour dimension. Several structures are batched by concatenating their atoms. For the Coulomb operator of molecular structures, the Coulomb matrix of a batch is block-diagonal, and the charge-equilibration systems of all structures in a batch are solved together.

5.3 Fused GPU kernels

Two custom kernels implement the parts of the model that dominated the profile of the PyTorch implementation.

Density kernel. The forward kernel evaluates Equation 6 with one thread block per atom and one thread per (b, k) pair. Tiles of ϕ, m and Y for consecutive neighbours are staged in shared memory, and each thread accumulates its 20 monomial components in registers. The per-edge tensor of size BK, which dominated the memory traffic of the PyTorch version,

is never stored. The backward kernel keeps the gradient of one atom's density in shared memory, forms per edge $T_{bk} = \sum_x G_{x,bk} Y_x$ once and reuses it for the gradients with respect to ϕ, m and Y. It is organised in three tiled phases with padding against shared-memory bank conflicts.

Node-algebra kernel. One thread per (atom, channel) evaluates all nine invariant families of Equation 13 and the gated-product paths of Equation 14 directly from symmetric storage, holding the full tensors in registers. Its backward pass folds the gradients of the full tensors back onto the symmetric components.

Both kernels support first derivatives only. They are therefore used for inference and molecular dynamics, where forces need one backward pass, whereas training, which needs second derivatives, uses the PyTorch path. The two paths compute the same function to within 1.4×10^{-6} (relative, float32). For inference, the learned radial functions can also be tabulated per element pair (4096 points, cubic Hermite interpolation), and the complete energy-and-force evaluation can be captured as a CUDA graph to remove kernel-launch overhead for small systems.

5.4 Verification

The test suite checks, in float64: invariance of E and equivariance of $\boldsymbol{F}$ under rotations, reflections, translations and permutations (to 10^{-9}); forces against finite differences (10^{-6} relative), including through the charge-equilibration solve and Ewald sums; finite second derivatives, which force training requires; C^1 continuity of energy and forces at r_c; lattice-translation invariance and exact supercell extensivity for periodic cells; the NaCl Madelung energy (2×10^{-6} relative for two Ewald splittings); SPME against exact Ewald (3×10^{-4} relative in energy, with the expected h^4 convergence in the grid spacing h); the transport maps against explicit per-edge evaluation; and the fused kernels against the PyTorch path (outputs and all gradients to $\leq 1.4 \times 10^{-6}$ relative).

6 Computational details

6.1 Data sets

rMD17[28] (PBE/def2-SVP). We use the official split 1: the 1000 training indices of split 1 (file `index_train_01`) are shuffled with a fixed seed into 950 training and 50 validation structures, and the 1000 test indices (`index_test_01`) form the test set, which is used only for final evaluation. Energy errors are for total energies of the molecule (meV per structure), force errors per Cartesian component (meV/Å). For the long-range benchmarks of Ko et al.[36] (FHI-aims, PBE, "light" settings), $Ag_3^{\pm}$ (11,013 structures) and $C_{10}H_2/C_{10}H_3^+$ (10,019) are shuffled with a fixed seed into 2000 training, 200 validation and 1000 test structures; these models were trained for 60 epochs with batch size 16 and peak learning rate 5×10^{-3}, and energies are reported per atom.

6.2 Baseline models

MACE[24] (mace-torch 0.3.16, e3nn 0.4.4) with 64×0e+64×1o (MACE-small, 241k parameters) and 128×0e+128×1o (MACE-medium, 752k) hidden irreps, correlation order 3, $l_{\max} = 3$, two layers, $r_c = 5$ Å, eight Bessel functions, trained with the same split, batch size, epoch budget, learning rate 0.01 (MACE default), AMSGrad, EMA 0.99 and MACE's two-stage energy/force weighting (1/1000, then 1000/100), in float32, and evaluated with the same script as FACE. MACE was run through its standard e3nn path; the NVIDIA cuEquivariance kernels, which accelerate MACE, are not available on the Windows system used, so our MACE timings are those of the reference implementation. Published values for MACE, Allegro, NequIP and others come from longer schedules, float64 and other hardware.

6.3 Model configurations

Table 4 lists the hyperparameters. The flagship has 1.56 M parameters, of which 0.06 M belong to optional, unused gates; FACE-CP has 1.31 M ($C = 96$) or 3.36 M ($C = 160$).

Parameter counts are large compared with MACE because the per-atom channel maps are cheap; cost is governed by the per-edge work (Section 3.7).

6.4 Timing protocol

One NVIDIA RTX 2000 Ada (16 GB), float32, energy and forces, periodic cubic boxes of randomly oriented aspirin molecules on a grid of spacing 5.97 Å (1.4 $\mathrm{g\,cm^{-3}}$; 1,344 atoms unless stated), neighbour list excluded for all models. Ratios are measured by interleaving all models in repeated rounds and taking medians, which makes them robust to background load; absolute idle-GPU numbers are given where measured. A process-level guard caps GPU memory at 12 GB.

6.5 Molecular dynamics

NVE dynamics use velocity Verlet with a 0.5 fs time step; Langevin dynamics use the ASE integrator with a friction of 0.01 $\mathrm{fs^{-1}}$ and the centre of mass fixed by a holonomic constraint. Initial velocities are drawn from a Maxwell–Boltzmann distribution with zero total momentum and angular momentum.

Table 4: Hyperparameters of the main configurations.

	flagship	FACE-CP
node width C	96	96 or 160
radial basis B / message rank K	16 / 32	16 / C (CP)
directional and transport channels K_d	16	16
cutoff r_c	4.5 Å	5.0 Å
message-passing steps T	2	2
radial MLP	$8 \to 64 \to 64 \to B$	$8 \to 64 \to 64 \to B + 4C$
readout MLP	$10C \to 2C \to 2C \to C + 1$	same
parameters	1.56 M	1.31 M / 3.36 M
optimiser	AMSGrad, weight decay 0, gradient clipping 10	
learning rate	one-cycle, peak 0.04, 5% warm-up, divisors 10 and 100	
batch size, EMA	5, 0.99 (parameters and buffers)	
epochs	60 (equal-budget comparison) or 400 (long training)	
loss weights (w_E, w_F)	(1, 1000), then (1000, 100) for the final 25% of epochs	

7 Results

The main findings, detailed in the subsections below, are:

- at equal training budgets, FACE is as accurate as MACE or more accurate, at 5.7 times (MACE-small) to 10.9 times (MACE-medium) lower inference cost and with about four times less GPU memory;
- widening FACE barely increases its cost, as predicted by the cost model, whereas widening MACE does;
- with little training data, FACE is competitive with NequIP and ACE, but its learning curve is slightly less steep than that of MACE;
- unit parameterisation removes the training divergences that affected earlier versions;
- uncertainty-gated distillation lowers the force error of a single model by 23–24% on aspirin, ethanol and malonaldehyde, and brings it to within 2–3% of the published MACE accuracy on aspirin;
- ensemble spreads are useful error indicators, and the models run stably in molecular dynamics and reproduce the thermal structure of the reference data;
- latent-charge electrostatics capture non-local charge effects in the long-range benchmarks, but the latent charges are not reliable atomic charges.

7.1 Accuracy

This subsection compares the accuracy of FACE with MACE at equal training budgets, identifies the components that matter, and follows the models through long training, ensembling and distillation.

7.1.1 Equal training budget

FACE *is at least as accurate as MACE at equal training budget.* Table 5 compares the flagship with MACE-small on five rMD17 molecules after 60 epochs. FACE is more accurate on aspirin, malonaldehyde, uracil and toluene and statistically tied on ethanol, where three seeds were run for both models (single 60-epoch runs vary by 20–30% between seeds). On aspirin, FACE also matches MACE-medium (8.5/18.3 vs 9.1/17.8 meV, meV/Å) while being 10.9 times faster (Figure 5a, Table 8). These 60-epoch errors are far from converged: published values for the same molecules after full training (Table S9) are 3–8 times lower. Converged training of FACE is reported below for aspirin and ethanol.

Table 5: rMD17, 60 epochs, identical splits: energy MAE (meV) / force MAE (meV/Å) on 1000 test structures. The flagship is 5.7× faster than MACE-small at inference.

molecule	FACE flagship	MACE-small	
aspirin	**8.45 / 18.3**	10.6 / 19.5	
ethanol (3 seeds, mean)	3.29 / 14.5	3.16 / 13.7	tie within seed spread
malonaldehyde	**6.28 / 16.9**	6.91 / 18.1	
uracil	**1.95 / 11.1**	2.40 / 12.8	
toluene	**1.46 / 6.82**	1.72 / 7.85	

7.1.2 Ablations

Message passing, higher-order equivariant messages and the rank of the edge factorisation matter most. Table 6 summarises the effect of the model components on aspirin. The upper part follows the development of the architecture at a fixed 60-epoch budget; the full sequence, including rejected variants, is given in Table S1. Because single 60-epoch runs vary by 20–30% between seeds, only larger differences are interpreted.

Three components carry most of the accuracy. Removing the second message-passing step doubles both errors, so the semi-local information that message passing adds is essential even for an isolated 21-atom molecule. Linear tensor transport alone changed the errors by only 13% and 9% at $C = 64$ (also with a higher learning rate), which is within the seed spread.

In these early, channel-wise models the edge rank equals C, and widening to $C = 96$ reduced the errors by a further 25% and 15%. The cleanest comparison is the addition of gated equivariant products at fixed width and learning rate, which reduced the errors by 26% and 21%. The second step therefore benefits most from equivariant information of higher than linear order in the first-step moments. The factorisation of the edge density cost accuracy at first (10.2/20.5 for the factorised model against 6.8/16.7 for the channel-wise model at the same C), in exchange for a 1.7-fold speed-up. Unit parameterisation, which allowed a four times higher learning rate, and the subsequent tuning recovered most of this loss (8.45/18.3), and the CP variant matches the channel-wise model (7.2/16.4).

The width experiments locate the bottleneck. Widening the nodes from $C = 96$ to 128 did not help (11.0/22.0), whereas a larger radial basis ($B = 32$: 8.8/18.1) or more directional and transport channels ($K_d = 48$: 8.6/18.0) did. The limiting factor is therefore the rank of the edge factorisation, in agreement with the discussion of Equation 8, not the node width.

The lower part of Table 6 lists variants tested with 400-epoch training, where seed variation is about 3%. Neither rank-3 transport nor element-dependent product gates improved the force error beyond the seed spread, and a polynomial (linear per-step) readout was clearly worse. A four times larger model (FACE-L, 6.3 M parameters, $C = 192$, $B = 32$, $K_d = 32$) reached the same accuracy as FACE-CP with 1.3 M parameters (2.90/8.72 vs 2.88/8.73). At this data size, model capacity is therefore not the limiting factor; the remaining error is generalisation, which motivates the distillation of Section 7.1.5.

7.1.3 Long training, ensembles and uncertainty

Longer training helps, ensembles help further, and the ensemble spread indicates where errors are large. With 400 epochs, single flagship models reach 3.0–3.4 meV and 8.8–9.6 meV/Å on aspirin (five models, three of them with element-dependent product gates; Figure S1). Validation errors plateau while training errors keep falling, i.e. the remaining error is generalisation rather than optimisation. RMSE values for all final models are listed in Table S6.

Table 6: Ablations on aspirin: energy MAE (meV) / force MAE (meV/Å), 1000 test structures.

configuration	E / F	note
60 epochs (seed spread 20–30%)		
scalar messages only (v0)	16.4 / 30.5	
+ directional invariants, element-pair radial (v1)	14.1 / 27.3	
+ tensor transport, $C = 64$ (v2)	12.2 / 25.0	lr 0.005 → 0.01
$C = 96$	9.2 / 21.2	
+ gated equivariant products, $C = 96$ (v3)	6.8 / 16.7	channel-wise edges
factorised edges, $B = 16$, $K = 32$ (v4)	10.2 / 20.5	1.7× faster than v3
single message-passing step	20.3 / 42.5	
$C = 128$	11.0 / 22.0	no gain
$B = 32$ / $K_d = 48$	8.8 / 18.1; 8.6 / 18.0	edge rank limiting
layer norm → learned affine	17.9 / 33.7	(v1)
+ unit parameterisation, lr 0.04	8.9 / 17.6	
flagship ($r_c = 4.5$ Å)	8.45 / 18.3	
FACE-CP	7.2 / 16.4	
400 epochs (seed spread about 3%)		
flagship, five models	2.97–3.42 / 8.82–9.60	
FACE-CP	2.88 / 8.73	
FACE-L[a] (6.3 M parameters)	2.90 / 8.72	no gain
rank-3 transport (two seeds)	2.91 / 8.30; 3.24 / 9.17	within spread
element-dependent product gates	2.97 / 8.82	within spread
polynomial readout	3.59 / 9.75	worse

[a] $C = 192$, $B = 32$, $K_d = 32$.

The five flagship models form an ensemble with 2.73 meV and 7.09 meV/Å, at a total inference cost still below one MACE-small evaluation. Figure 2c shows that the ensemble spread is a useful error indicator: the Spearman rank correlation between the per-structure force spread and the per-structure force error of the ensemble mean is 0.84 (Pearson 0.84), and the 10% of test structures with the smallest spread have a mean force error of 4.0 meV/Å, compared with 12.1 meV/Å for the 10% with the largest spread. The spread is also calibrated in magnitude: its test-set mean (6.6 meV/Å) is close to the mean error (7.1 meV/Å). At the level of individual atoms the rank correlation is 0.73. The same spread predicts the errors of the best distilled single model (rank correlation 0.84), so a cheap ensemble of flagship models can serve as an uncertainty monitor for a more accurate production model.

Figure 2a,b shows the error distribution of the best single model (generation 4, below). The force errors are symmetric and narrow: the median absolute component error is 4.3 meV/Å, the 90th and 99th percentiles are 15.6 and 36.3 meV/Å, and the largest of 63,000 components is 177 meV/Å, compared with a standard deviation of the reference force components of 1.27 eV/Å. Errors are smallest for hydrogen (3.3 meV/Å) and larger for carbon (9.2) and oxygen (8.2), reflecting the stiffer and more varied environments of the heavy atoms. The median absolute energy error is 1.7 meV and the 99th percentile 9.5 meV.

7.1.4 Data efficiency

FACE *learns about as efficiently as NequIP, and less efficiently than MACE.* To test how the accuracy depends on the amount of training data, the flagship was trained on nested subsets of $N = 50$, 100, 250 and 500 of the 950 training structures, with the same validation and test sets, for 400 epochs or at least 20,000 optimiser steps, whichever is more. The $N = 500$ run was stopped by the memory guard of the training computer at epoch 380 of 400, after the validation force error had stopped changing (11.63–11.64 meV/Å over the last 50 epochs), and was evaluated at that point. Figure 3 compares the test errors with published values for MACE, NequIP and ACE at 50 and 1000 training structures.[24]

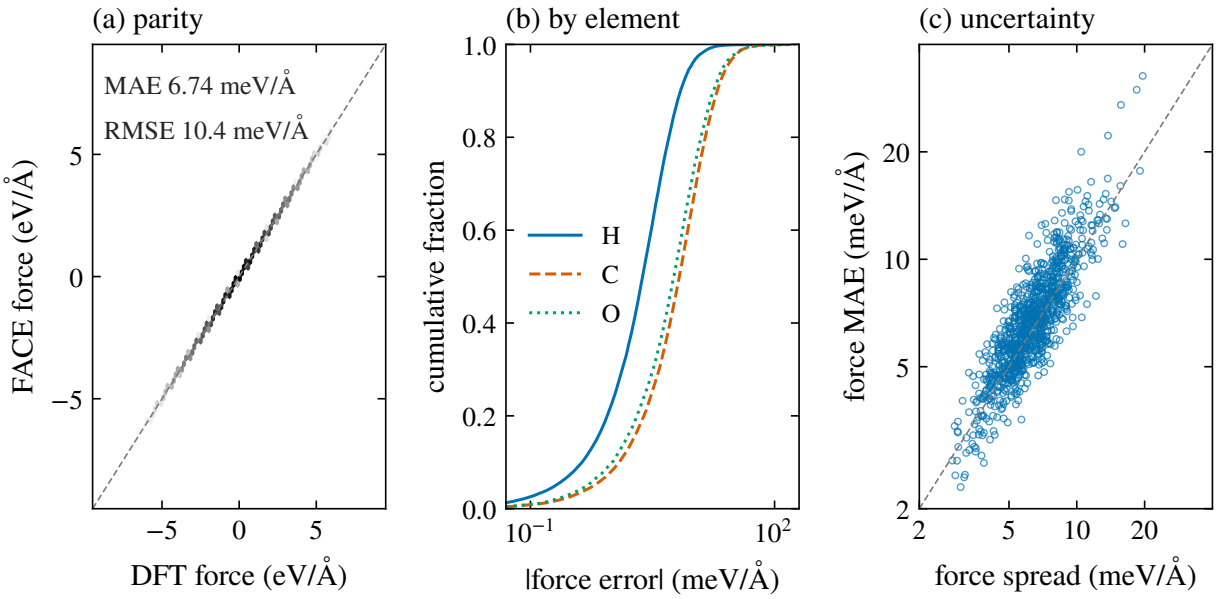


Figure 2: Error analysis on the aspirin test set (1000 structures). (a) Predicted versus reference force components of the best single model (generation-4 FACE-CP; logarithmic point density). (b) Cumulative distribution of absolute force-component errors by element. (c) Per-structure force error of the five-model flagship ensemble versus its force spread (Spearman 0.84); the dashed line marks equality.

The force error decreases from 50.3 meV/Å at $N = 50$ to 9.08 meV/Å at $N = 950$, following a power law with exponent 0.58 (energy: 0.67). With 50 structures, FACE (24.6 meV, 50.3 meV/Å) has a lower force error than NequIP (19.5, 52.0) and ACE (26.2, 63.8) but a higher one than MACE (17.0, 43.9); its energy error lies between those of NequIP and ACE. Between the two published points, the force errors of MACE and NequIP fall with slightly steeper exponents (0.64 and 0.63) than that of FACE, and that of linear ACE with a flatter one (0.43); the force-error gap to MACE accordingly grows from 15% at $N = 50$ to 38% at $N = 950$. The remaining gap to MACE at large N is therefore one of data efficiency rather than capacity, in agreement with the FACE-L result of Section 7.1.2; it is this gap that the distillation of Section 7.1.5 reduces. The published values come from longer training in double precision and are not strictly comparable; the values of FACE are listed in Table S13.

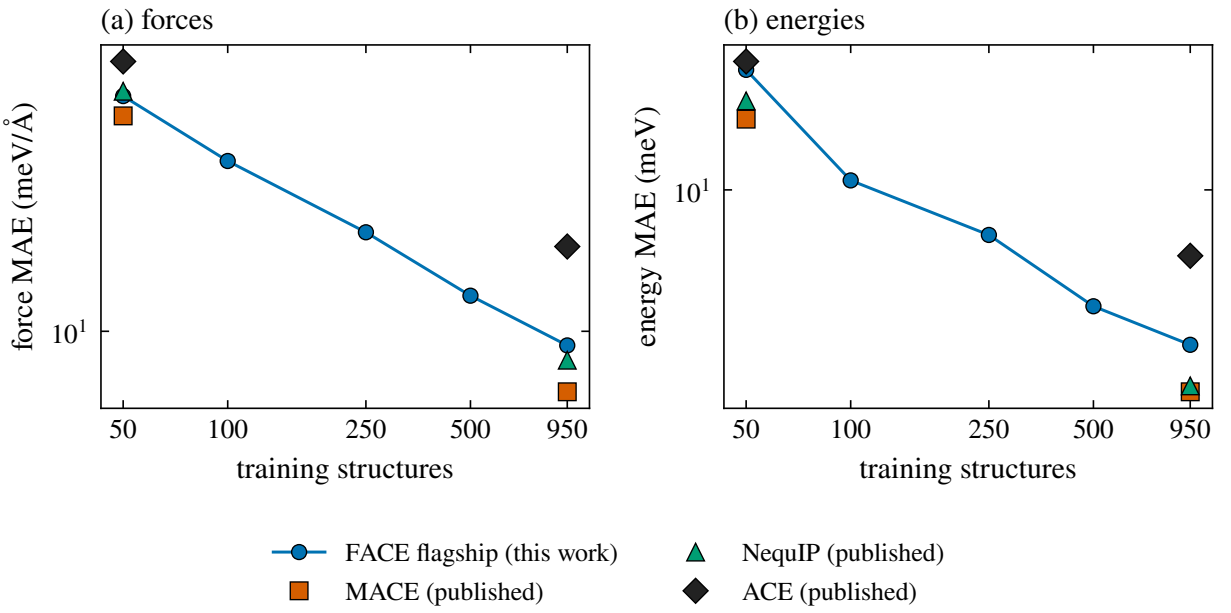


Figure 3: Test errors of the FACE flagship on rMD17 aspirin versus the number of training structures (nested subsets of the training set, the same 1000 test structures), compared with published values for MACE, NequIP and ACE at 50 and 1000 training structures (plotted at 950),[24] shown as points because only these two values are published. (a) Force MAE; (b) energy MAE.

7.1.5 Uncertainty-gated distillation

Distillation from thermal teacher dynamics brings a single model close to the best published accuracy. Figure 4a shows why the distillation data must be thermal. Random Gaussian displacements of 0.02–0.06 Å place structures where the teacher members disagree by 9.5–150 meV/Å, because uncorrelated displacements excite stiff modes far beyond thermal amplitude, whereas teacher MD at 500 K gives a spread of 6.9 meV/Å, close to the 5.5 meV/Å on the real test set; only 0.4–1.5% of MD snapshots are rejected by the gate. Figure 4b,c and Table 7 show the effect of iterated distillation on a single FACE-CP model. Each generation reduced the force error, from 8.73 meV/Å without distillation to 7.46, 7.04 and 6.97 meV/Å in generations 1–3. A wider student was needed once the teacher (11 models, 2.55/6.53) surpassed the published MACE accuracy, and the final energy-weighted fine-tune reduced the energy error by 9% at a 0.6% force cost. The distilled generation-4 model (2.26/6.74) is within 3% and 2% of the published MACE result and better on both errors than published NequIP, Allegro and TensorNet results. The flagship benefits from distillation as well (3.13/8.04 from 3.13/9.08) while keeping its 5.7-fold speed advantage. Supplying more teacher data at the same optimiser budget did not help (Table S5), which suggests that the gain comes from the smoother labels on thermal configurations near the training data rather than from the amount of data.

To test whether the protocol transfers without tuning, it was applied unchanged to malonaldehyde, with three FACE-CP teachers ($C = 96$, 400 epochs; single-model errors 1.12–1.22 meV and 7.13–7.38 meV/Å; ensemble 0.98/6.02) and a $C = 160$ student. The gate rejected 79 of 19,000 teacher-MD snapshots. After the DFT-only fine-tune, the single student reaches 0.91 meV and 5.46 meV/Å, 24% lower in force error than its teachers and better than the teacher ensemble; its force error is lower than published BOTNet (5.8) but higher than published NequIP (5.1), MACE (4.1) and Allegro (3.6).[18,24] Across the three molecules, distillation lowers the force error of a single FACE-CP model by a similar amount: 23% for aspirin (8.73 to 6.74 meV/Å), 23% for ethanol (4.84 to 3.72 meV/Å, Section 7.1.6) and 24%

for malonaldehyde.

Table 7: Aspirin, single models, 1000 test structures (meV, meV/Å). Published values are from different protocols.

model	E MAE	F MAE
FACE-CP, DFT data only, 400 epochs	2.88	8.73
generation 1 (7-model teacher), + DFT fine-tune	2.71	7.46
generation 2 (9-model teacher), + fine-tune	2.69	7.04
generation 3 (10-model teacher), + fine-tune	2.50	6.97
generation 4, $C = 160$ student (11-model teacher), + fine-tune	2.48	6.70
generation 4, + energy-weighted fine-tune	**2.26**	**6.74**
FACE flagship, distilled + fine-tune (5.7× MACE-small speed)	3.13	8.04
published TensorNet[31]	2.4	8.9
published NequIP[17]	2.3	8.2
published Allegro[18]	2.3	7.3
published MACE[24]	2.2	6.6
published ViSNet[a 21]	1.93	6.59
published QuinNet[a 22]	2.11	6.20
published GotenNet[a 23]	1.55	5.66

[a] converted from kcal/mol.

7.1.6 Ethanol

On ethanol the gap to published values is a matter of training protocol, not of the model. On ethanol, FACE trails the published best values considerably more than on aspirin (single models 0.74–0.78/ 4.7–4.9 vs 0.4/2.1). A control resolves this: MACE-small trained with the identical 400-epoch protocol reaches 1.11/6.76, so FACE is about 30% more accurate than MACE-small at equal training, and the gap to the published number reflects the much larger float64 MACE configuration and its longer training. Distillation improves the single model further. A first generation with a three-model teacher gives 0.68/4.22. A second generation, whose four-model teacher includes the first student (teacher error 0.60/3.73) and whose student is a wider FACE-CP with $C = 160$, gives 0.62/3.72 after the DFT-only fine-tune: the single student now matches its teacher, and its force error is 45% below that of MACE-small under the same protocol. Unlike for aspirin, an additional energy-weighted

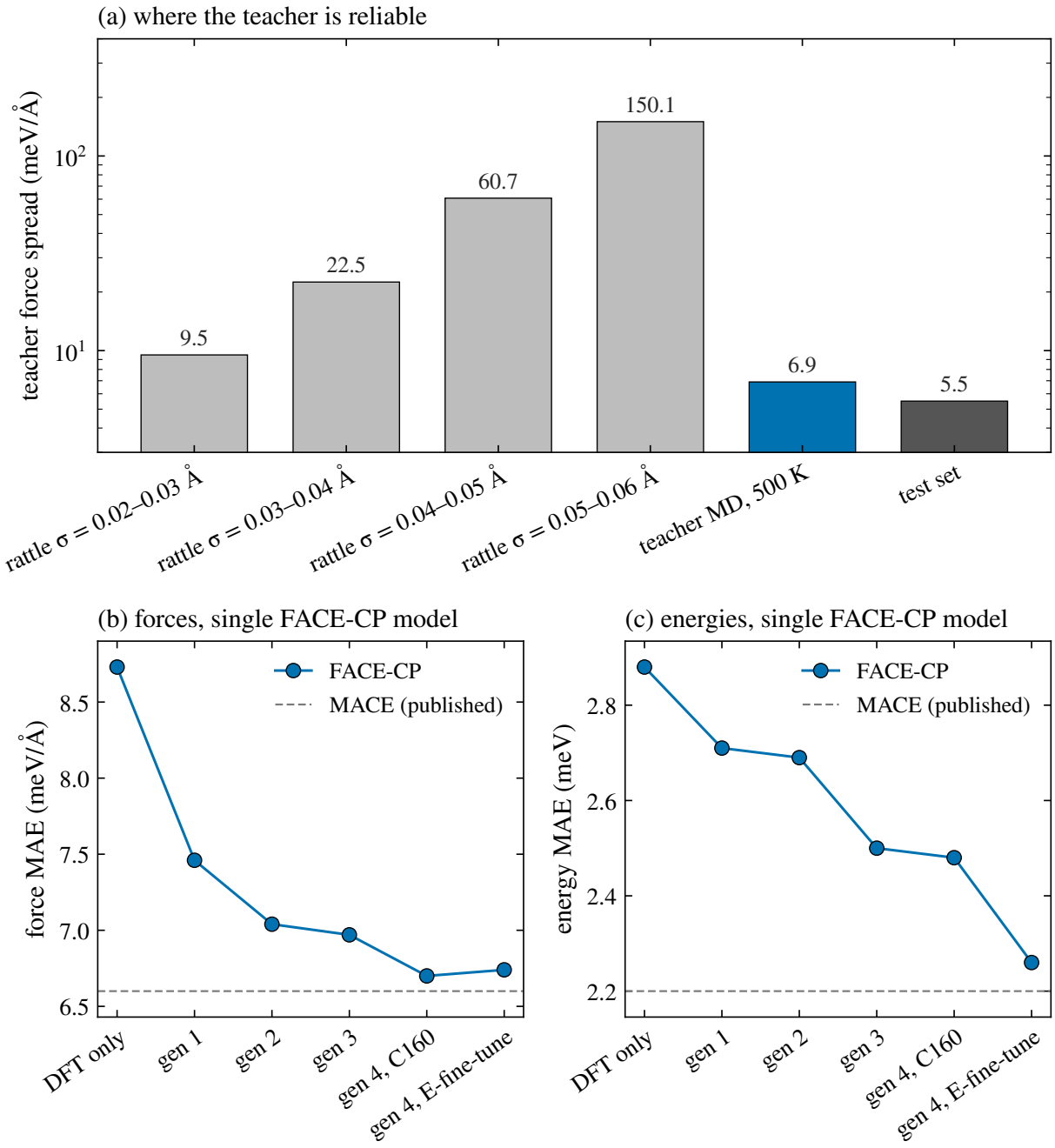


Figure 4: (a) Teacher force spread on randomly rattled structures, on teacher-MD snapshots and on the real test set. (b, c) Force and energy MAE of a single FACE-CP model across distillation generations.

fine-tune did not lower the ethanol energy error (0.64/3.68). The gap to the published MACE value (0.4/2.1) remains about 1.8-fold in the forces. Ethanol has a low-barrier methyl and hydroxyl rotation, and its small size gives only eight neighbours per atom on average; both make it more sensitive to the description of torsions than aspirin.

7.2 Computational cost

7.2.1 Inference cost

FACE *is several times cheaper than MACE at inference, but not in training.* Table 8 lists the inference cost at 1,344 atoms. The flagship evaluates energy and forces in 33.3 μs per atom, 5.7 times faster than MACE-small and 10.9 times faster than MACE-medium. The fastest configuration ($C = 64$) is 6.3 times faster than MACE-small but less accurate on aspirin, and FACE-CP trades a factor of two in speed for higher accuracy. The published MACE configuration with 256 channels and $L = 2$ needs 1.6 ms per atom at 168 atoms and does not fit in 12 GB at 1,344 atoms.

Table 8: Inference cost (energy + forces, float32, RTX 2000 Ada, 1,344 atoms) and training-step cost (batch of five aspirin molecules, including second derivatives and the optimiser step, relative to MACE-small). Idle-GPU inference values measured directly except those marked [a], derived from interleaved ratios to MACE-small under identical load. "vs MACE-s/-m": inference speed-up relative to MACE-small/-medium; "train step": training-step time relative to MACE-small.

model	μs/atom	vs MACE-s	vs MACE-m	train step
MACE-medium (128 ch)	361.5	0.53×	1.0×	1.35
MACE-small (64 ch)	190.9	1.0×	1.9×	1.00
FACE flagship	**33.3**	**5.7**×	**10.9**×	0.96
FACE $C = 64$	30.4	6.3×	11.9×	–
FACE-CP $C = 96$	≈66[a]	2.9×	5.5×	0.95
FACE-CP $C = 160$	≈97[a]	2.0×	3.7×	0.98
MACE 256 ch, $L = 2$	1634[b]	–	–	–

[b] measured at 168 atoms; out of memory at 1,344 atoms.

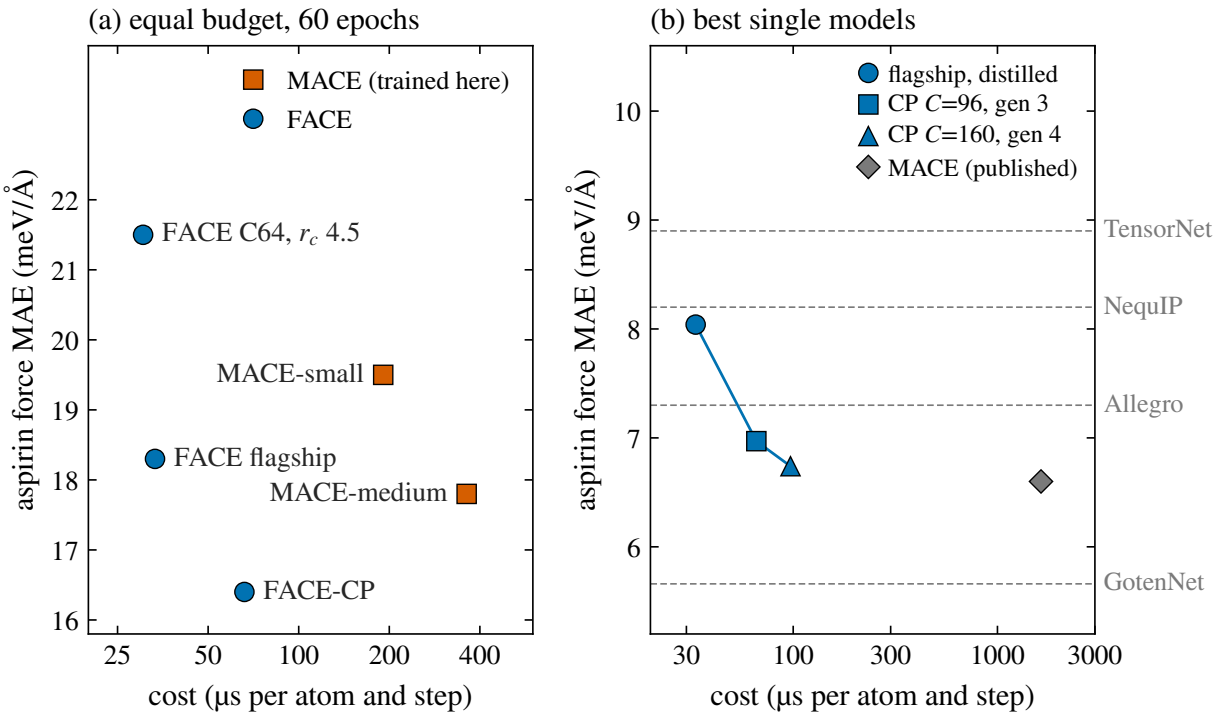


Figure 5: Accuracy versus inference cost (energy and forces, μs per atom and step) on aspirin. (a) Equal 60-epoch budget, all models trained here. (b) Best single models after long training and distillation versus published long-training results (dashed lines). The published MACE configuration has 256 channels and $L = 2$; its cost was measured on our GPU at 168 atoms because it exceeds the memory limit at 1,344 atoms.

7.2.2 Where the cost goes

Profiling guided a sequence of exact, output-identical optimisations: replacing multi-operand einsums by broadcast contractions (9–19× on those operations), performing the expansion from symmetric to full tensors as a general matrix multiplication (GEMM) with a constant 0/1 matrix, pooling with one GEMM per rank on the small monomial array instead of slicing the large edge tensor, the fused density kernel (forward 2.1× faster than PyTorch; backward 150 ms → 40 ms), the fused node-algebra kernel (6.3×), the constant-map transport and spline tables for narrow radial bases. The flagship went from 1.6× to 5.7× the speed of MACE-small (Table S3). Two negative results are worth recording: mixed precision gave only 6–12% because the remaining work is in memory-bound elementwise kernels, and spline-compiled radial functions are slower than a small GEMM for wide radial heads. CUDA-graph capture of the full energy and force evaluation, including the custom kernels, reduces launch overhead by 2.4× at 168 atoms but has no effect at 4.5k atoms.

7.2.3 System-size dependence

Figure 6 shows that the advantage depends on system size (Table S7): the flagship is 2.3× faster than MACE-small at 168 atoms, 4.7× at 567, 5.6× at 1,344 and 5.4× at 2,625 atoms (2.4× to 3.2× for FACE-CP). Small systems are limited by kernel-launch latency rather than arithmetic, and CUDA-graph replay recovers a factor of 2.4 at 168 atoms. Above about 1,000 atoms the ratio is constant, because both models then scale linearly with the number of atoms.

7.2.4 Width and memory

Figure 7a tests the central claim of the cost model directly. At 1,344 atoms, widening the flagship eightfold, from $C = 32$ to $C = 256$, increases its cost only 1.67-fold, from 0.16 to 0.27 of the cost of MACE-small. Quadrupling the MACE width from 32 to 128 channels increases its cost 3.25-fold. FACE-CP lies in between (1.9-fold from $C = 64$ to 160), as

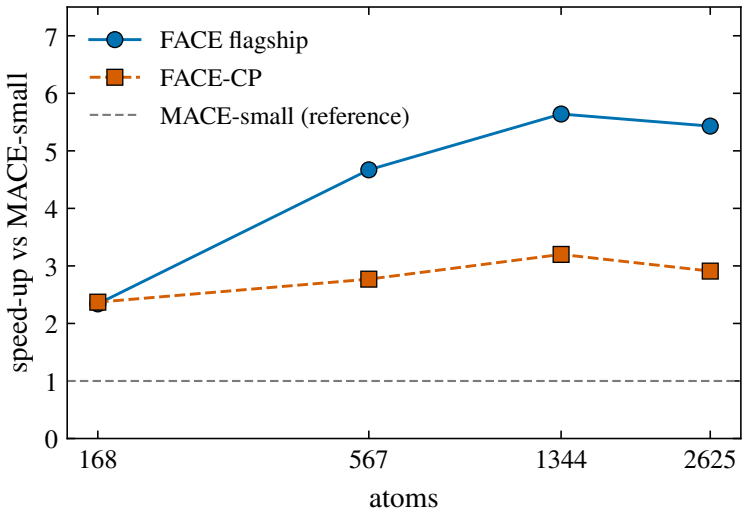


Figure 6: Inference speed-up relative to MACE-small as a function of system size (periodic aspirin boxes, energy and forces, float32; medians of interleaved timing rounds).

expected, because its edge work grows with C. Peak GPU memory for energy and forces (Figure 7b) grows linearly with the number of atoms for all models, at about 0.5 MB per atom for the flagship, 1.3 MB for FACE-CP and 2.0 and 3.9 MB for MACE with 64 and 128 channels. The flagship needs 3.5–3.9 times less memory than MACE-small at every size. At 4,536 atoms it uses 2.3 GB, whereas the 128-channel MACE no longer fits within the 12 GB limit. Extrapolated linearly, a 12 GB budget holds about 23,000 atoms with the flagship and about 6,000 with MACE-small, so the memory advantage extends the attainable system size on a single GPU by a similar factor as the speed advantage. For molecular dynamics, the flagship's 33.3 μs per atom and step corresponds to about 1 ns per day for 1,344 atoms with a 0.5 fs time step on this consumer GPU, excluding the neighbour list.

7.2.5 Training cost

Training is *not* cheaper. A training step on five aspirin molecules, which requires second derivatives that the fused kernels do not cover, costs 0.95–0.98 of a MACE-small step and 0.7 of a MACE-medium step (Table 8 and Table S8). A 400-epoch flagship run took 3.1 h on the same GPU. The distilled models of Table 7 carry a further training overhead: generation

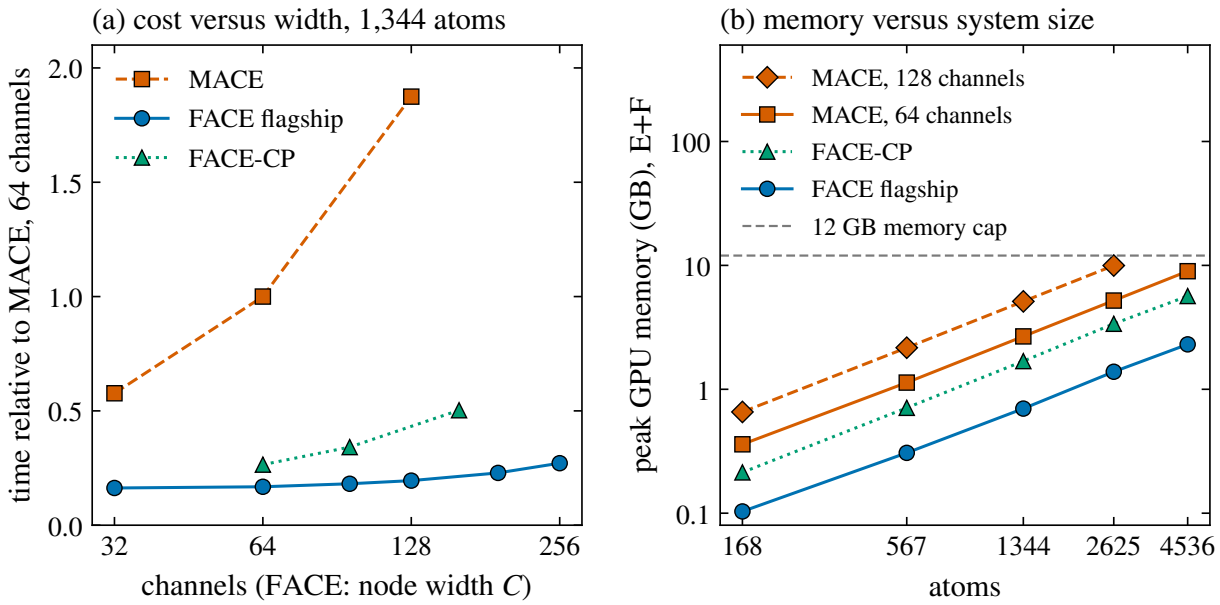


Figure 7: (a) Inference cost versus model width at 1,344 atoms, relative to MACE with 64 channels measured in the same interleaved rounds; for FACE the width is the node width C. (b) Peak GPU memory for one energy and force evaluation versus system size; MACE with 128 channels exceeds the 12 GB cap at 4,536 atoms. Periodic aspirin boxes, float32.

4 depends on seven 400-epoch teacher runs and four earlier students and their fine-tunes, in total about 1.1×10^6 optimiser steps, or roughly 14 times the training compute of a single 400-epoch model. Distillation therefore trades one-off training compute for inference cost and accuracy.

7.3 Training stability

The divergences were caused by the weight parameterisation, not by the architecture. Early versions diverged on ethanol (NaN gradients from step 16 on every batch, with all weights finite). The inputs of the invariant layer normalisation had grown to 10^7 and 3×10^{20}, overflowing in the second derivative. Layer normalisation is invariant to the overall scale of its input, so the optimiser can enlarge the moments without changing the loss, and this scale direction reweights the body orders, because a ν-body invariant grows as the $(\nu-1)$-th power of the moment scale. Normalising the moments per atom stabilised training but doubled the

aspirin error, and data-calibrated fixed scales without layer normalisation diverged within 2–33 steps (Table S2). A controlled probe at a constant learning rate located the cause in the weight parameterisation: with the default parameterisation the relative change per step of the 1024-wide edge maps is about 30% (Section 4). With unit parameterisation the same architecture trained for 1500 steps at learning rate 0.01 on ethanol without a single non-finite value (default parameterisation: first non-finite step 85), and the optimal peak learning rate moved from 0.01 to 0.04. One of three ethanol seeds still diverged at learning rate 0.04 in a 60-epoch schedule; the recovery mechanism guards against this.

7.4 Long-range interactions and charge

7.4.1 Benchmarks of Ko et al.

Latent-charge electrostatics capture non-local charge effects that a local model cannot. Table 9 summarises the Ko et al. benchmarks. For $C_{10}H_2/C_{10}H_3^+$ with the default cutoff, two message-passing steps reach about 10 Å, comparable to the molecular length, and local and charge-equilibrated models perform equally. With $r_c = 3$ Å (6 Å receptive field, shorter than the 13 Å chain), charge equilibration halves both errors: the protonation state at one end of the chain changes the charge distribution at the other, which only a non-local model can represent. For $Ag_3^{\pm}$, latent charges with the total charge as a constraint reduce the energy error about 100-fold relative to the best local model. The two charge states were sampled by separate MD and occupy mostly different geometries, so this set is not a same-geometry test; a local model without equivariant saturation was numerically unstable on it (test error about 1.2 eV/atom). These trends agree with the original study,[36] which reports energy errors of 0.6–2.0 eV/atom for a local second-generation network and 0.3–1.2 meV/atom for the fourth-generation network on the DFT-optimised $Ag_3^{\pm}$ structures; because those values refer to optimised structures rather than to a test set, we do not compare them numerically with Table 9.

7.4.2 Are the latent charges physical?

The latent charges are never fitted to reference charges; they are intermediate variables determined by the energy and forces. Figure 8 compares them with the Hirshfeld charges of the reference calculations[36] on the 1000 test structures of each set. Over all atoms, the correlation is high for $Ag_3^{\pm}$ (Pearson 0.96) and moderate for the carbon chains (0.68), but the overall value is dominated by the total charge. Within each charge state the picture is mixed. For Ag_3^+ the latent charges follow the reference closely (correlation 0.89, mean absolute deviation 0.011 e), and for the neutral $C_{10}H_2$ they reproduce the atom-resolved Hirshfeld pattern (0.84). For Ag_3^-, however, the model distributes the excess electron almost evenly (standard deviation 0.022 e against 0.112 e in the reference) and the within-state correlation is negative (-0.82); for $C_{10}H_3^+$ it is absent (0.03). The latent charges therefore carry the total-charge information that the energy requires, and in some systems they resemble a population analysis, but they are not a substitute for one. Where physically meaningful charges or dipoles are needed, they should be added to the loss, as in the fourth-generation networks;[36] the operator interface allows this without changes to the model.

7.4.3 Cost of the physics operators

The dispersion and ZBL operators were verified for correctness (symmetry, forces, periodicity) and cost only; none of the data sets used here probes van der Waals binding or close nuclear contacts, so their benefit to accuracy remains to be shown. Their cost was measured against the local flagship with fused kernels, interleaving all variants in the same timing rounds (Table S10). All three operators together add 23% to the cost at 4,536 atoms, 28% at 2,625, 44% at 1,344 and 81% at 567 atoms. Dispersion adds 9–12% at every size, because its 10 Å pair list is several times longer than the model's neighbour list, and ZBL adds 2–8%. The electrostatic operator dominates in small systems: it adds 62% at 567 atoms but only 12% at 4,536 atoms, because the charge-grid and Fourier-transform steps of SPME carry a fixed cost that is large compared with the fast local model when there are few atoms. The

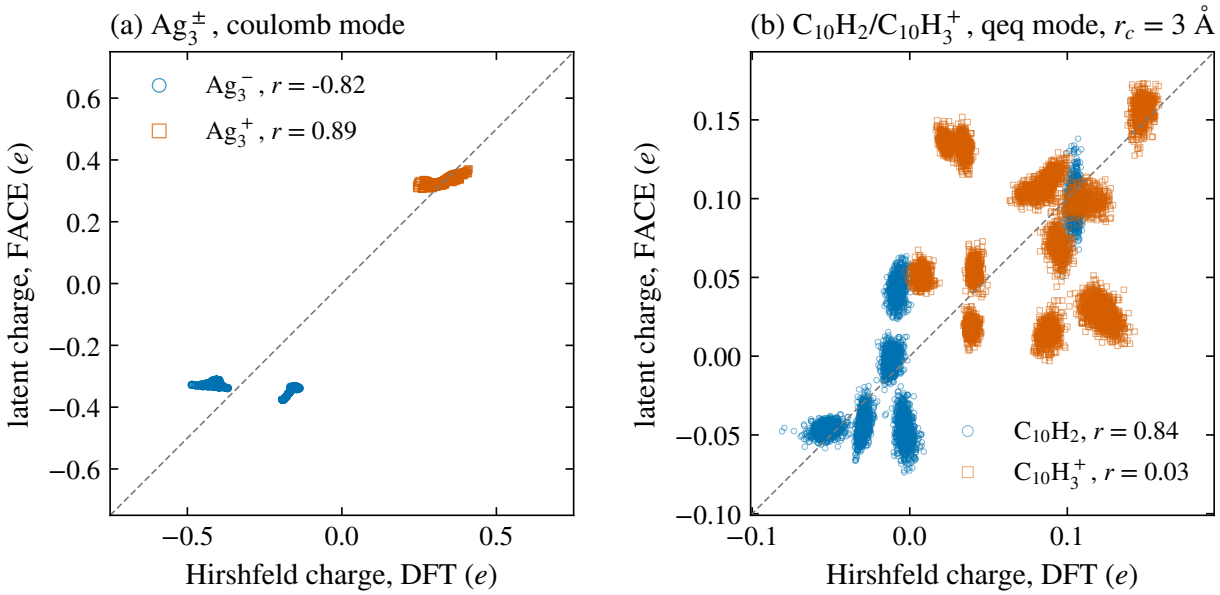


Figure 8: Latent charges predicted by FACE versus Hirshfeld charges of the reference DFT calculations, which were not used in training, for the 1000 test structures of (a) $Ag_3^{\pm}$ and (b) $C_{10}H_2/C_{10}H_3^+$ ($r_c = 3$ Å). r: Pearson correlation within each charge state. Dashed line: equality.

operators are therefore affordable for large systems, but a substantial fraction of the cost for small ones, and they should be switched on only where the physics requires them.

Table 9: Long-range benchmarks (Ko et al.), 1000 test structures: energy MAE (meV/atom) / force MAE (meV/Å).

system	local FACE	with latent charges
$C_{10}H_2/C_{10}H_3^+$, $r_c = 5$ Å	0.33 / 12.6	0.36 / 14.5 (qeq)
$C_{10}H_2/C_{10}H_3^+$, $r_c = 3$ Å	0.71 / 32.8	**0.37** / **17.4** (qeq)
Ag_3^+/Ag_3^-	214 / 440[a]	**1.94** / **9.7** (coulomb)

[a] with equivariant saturation; without it training was numerically unstable.

7.5 Molecular dynamics

All models are stable in molecular dynamics and reproduce the thermal structure of the reference. Low test errors do not guarantee stable dynamics,[68,69] so all main models were run in NVE dynamics. All conserved energy: 20 ps at 500 K, drift 3×10^{-4} meV/atom/ps;

10 ps from 1000 K (mean 658 K), 9×10^{-4} meV/atom/ps; the flagship with fused kernels and spline-compiled radial functions, -0.004 meV/atom/ps; the distilled generation-3 model, 5 ps from 1000 K, -0.002 meV/atom/ps. No bonds broke in any run. This contrasts with direct-force models, for which unstable NVE dynamics have been reported.[69]

Figure 9 documents a longer test of the best distilled model. In 20 ps of NVE dynamics started at 500 K, the total energy fluctuates by 0.06 meV/atom (standard deviation) around a drift of -0.004 meV/atom/ps, while the potential energy fluctuates by 8.0 meV/atom. In 20 ps of Langevin dynamics at 500 K, the distributions of all 210 interatomic distances agree with those of the rMD17 reference data, which were sampled by ab initio MD at 500 K[27] (mean kinetic temperature after the first picosecond: 509 K). The overlap of the two normalised distance histograms is 0.97, and the mean bond lengths agree to within 0.01 Å (C–H +0.009, C–C +0.006, C–O -0.006, O–H +0.0004 Å), which is small compared with their thermal widths of 0.03–0.10 Å. The model therefore conserves energy in NVE dynamics and reproduces the thermal structural ensemble of the reference.

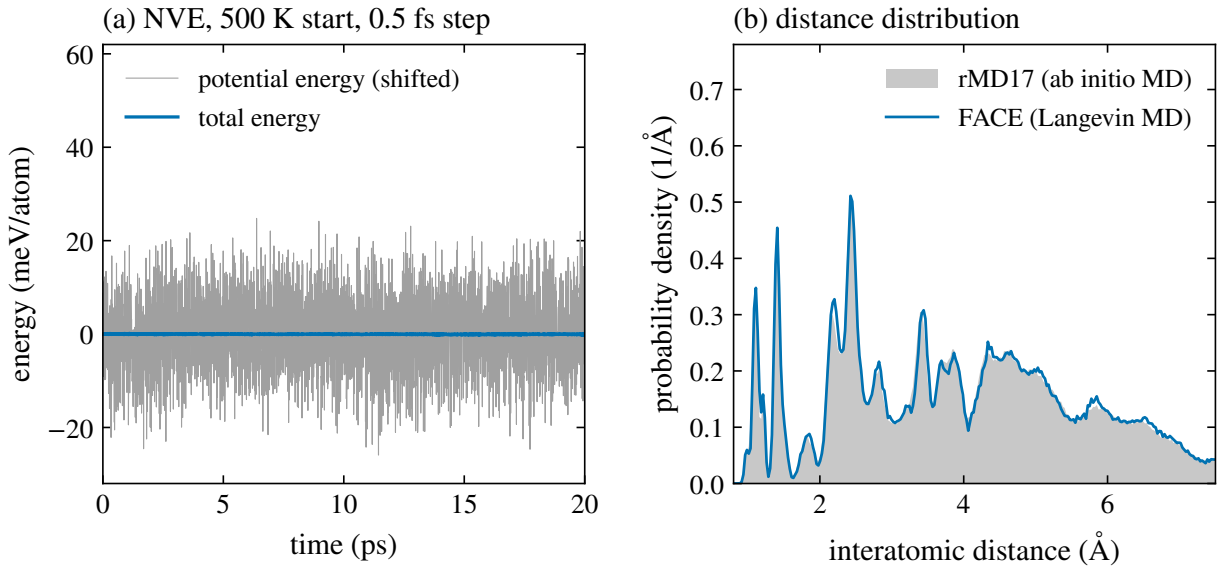


Figure 9: Molecular dynamics of aspirin with the best distilled model. (a) Total and potential energy per atom in 20 ps of NVE dynamics from 500 K (0.5 fs time step); the potential energy is shifted for display. (b) Distribution of all interatomic distances in Langevin dynamics at 500 K compared with the rMD17 reference data.

8 Discussion

8.1 Position among related models

FACE is closest to MTP and MACE. Like MTP, it uses Cartesian moment tensors of the bond directions; like MACE, it builds the density from learned neighbour features and forms body-ordered products per atom over two message-passing steps. Its differences are the factorisation of Equation 8, which moves the channel dimension from the edges to the atoms; the transport of equivariant neighbour tensors through a constant map instead of per-edge tensor products; and the evaluation of all products in symmetric Cartesian storage in fused kernels. TensorNet and CACE also use Cartesian tensors but keep channel-wide tensors on each edge, and ICTP replaces spherical by irreducible Cartesian tensor products, which are still evaluated on every edge. Allegro achieves its efficiency differently, by restricting messages to edges and avoiding atom-to-atom message passing, which helps parallel scaling; the factorisation in FACE is compatible with that design in principle. The CP variant shows how the factorisation interpolates between the two regimes: with a rank equal to the node width it recovers MACE's first-layer density and costs about twice as much as the flagship, still three times less than MACE-small.

The results define when each configuration is useful. The flagship is the choice for long simulations where inference cost dominates and an error of about 9 meV/Å after standard training (8 meV/Å after distillation) is acceptable. FACE-CP is about twice as expensive and reaches lower errors, in particular after distillation. The ablations indicate that the accuracy of the flagship is limited by the rank BK of the edge factorisation, so the most effective way to trade cost for accuracy is to enlarge B or K_d, not C.

8.2 Limitations

The theoretical foundation is not new; the novelty lies in the factorisation, the transport, the kernels, the parameterisation and the training protocol. rMD17 is a saturated single-

molecule benchmark. Materials, surfaces, reactive chemistry and extrapolation tests (e.g. 3BPA[70]) were not performed, nor were the NaCl and Au_2/MgO systems of Ko et al. All speed measurements use one consumer GPU in float32. The published MACE accuracy was obtained in float64, and MACE was timed without the cuEquivariance kernels that are available on Linux, so the speed ratios to MACE would shrink with those kernels; a comparison on multi-GPU or CPU hardware and in production MD codes is outstanding. The advantage concerns inference: training steps cost about as much as MACE-small, and a distilled model requires roughly an order of magnitude more training compute than a single model. Distillation was demonstrated on three molecules (aspirin, ethanol and malonaldehyde) and does not add information beyond the 950 reference structures; it transfers the ensemble's averaged knowledge onto thermally sampled configurations. FACE does not reach the most accurate published aspirin results (ViSNet 1.93/6.59, QuinNet 2.11/6.20, GotenNet 1.55/5.66 meV, meV/Å), whose cost we have not measured. Several comparisons rest on single training runs, whose spread we quantified (about 3% at 400 epochs, 20–30% at 60 epochs). The latent charges reproduce the total-charge partitioning but, without charge reference data in the loss, not reliably the atomic charges themselves (Figure 8). Finally, ensembles are more expensive than single models; the uncertainty they provide could instead be obtained from last-layer methods.[71]

8.3 Outlook

Three extensions follow directly from the results. First, fused kernels for the second-derivative pass would extend the inference advantage to training. Second, the physics operators should be tested where they matter: ionic materials, interfaces and molecular crystals, for which electrostatics and dispersion dominate the long-range energy. Third, the factorisation leaves the per-atom part of the model free to grow, so multi-element systems, where MACE-type models use element-dependent weights in the product basis, are a natural test of whether the edge rank or the chemical complexity limits accuracy.

9 Conclusions

FACE is a factorised Cartesian atomic cluster expansion in which the edge cost is independent of the feature width, tensor transport is a constant linear map, and the body-ordered algebra runs in fused kernels. Physics operators for electrostatics, dispersion and repulsion attach through one interface and add about 23% to the cost of a 4,500-atom system. Unit parameterisation removes the training divergences of this model class, and uncertainty-gated, iterated ensemble distillation with DFT-only fine-tuning brings a single model within 3% (energy) and 2% (forces) of the published MACE aspirin result, at an inference cost below that of a 64-channel MACE. The model is energy-conserving in long molecular dynamics and reproduces the thermal structure of the reference data, and ensembles of the fast configuration provide calibrated uncertainties. Natural next steps are fused kernels for the training backward pass, benchmarks on condensed-phase and reactive systems where the physics operators matter, and implementation in production molecular-dynamics codes.

Acknowledgement

This research received no specific grant from any funding agency in the public, commercial or not-for-profit sectors. We thank the developers of rMD17, MACE, e3nn and ASE, and Ko et al., for making their data and software openly available.

Supporting Information Available

The Supporting Information is available free of charge on the ACS Publications website: architecture development history and ablations; training-stability probe; kernel engineering; ensembles and seed variance; additional distillation results; energy and force RMSE values; inference cost versus system size; training-step cost; published reference values; cost of the physics operators, inference time versus width and GPU memory versus system size; learning-

curve values; hyperparameters; entries of the transport map; validation curves (PDF).

Author Information

Corresponding Author: Albert Linda – Department of Materials Science and Engineering, Indian Institute of Technology Kanpur, Kalyanpur, Kanpur 208016, Uttar Pradesh, India; ORCID 0000-0002-3723-1756; E-mail: albert.hzbn@gmail.com.

Author Contributions: Albert Linda conceived the study, designed and implemented the model, performed all calculations and wrote the manuscript.

Notes: The author declares no competing financial interest.

Data and Software Availability

The FACE source code (Python/PyTorch with CUDA kernels compiled at run time through CuPy/NVRTC), the test suite, all training logs, trained checkpoints, generated distillation data sets and the scripts that produce every table and figure are publicly available under the MIT licence at `https://github.com/albert-hzbn/face-mlip`. Software versions: Python 3.12, PyTorch 2.6.0 (CUDA 12.4), CuPy 14.2, ASE 3.29,[67] mace-torch 0.3.16, e3nn 0.4.4. The rMD17 data set is available from figshare (article 12672038); the data of Ko et al. from the Materials Cloud Archive (record bk6kw-4b895).

Use of AI Tools

An AI system (Claude Opus 5.5, Anthropic) assisted with the literature survey, the writing and debugging of the software, running and analysing the experiments, and drafting and editing the manuscript and figures. All numerical results were produced by the code in the accompanying repository and can be reproduced from it. The author reviewed and edited all content and takes full responsibility for the work; the AI system is not an author.

References

(1) Behler, J.; Parrinello, M. Generalized Neural-Network Representation of High-Dimensional Potential-Energy Surfaces. *Phys. Rev. Lett.* **2007**, *98*, 146401.

(2) Bartók, A. P.; Payne, M. C.; Kondor, R.; Csányi, G. Gaussian Approximation Potentials: The Accuracy of Quantum Mechanics, without the Electrons. *Phys. Rev. Lett.* **2010**, *104*, 136403.

(3) Unke, O. T.; Chmiela, S.; Sauceda, H. E.; Gastegger, M.; Poltavsky, I.; Schütt, K. T.; Tkatchenko, A.; Müller, K.-R. Machine Learning Force Fields. *Chem. Rev.* **2021**, *121*, 10142–10186.

(4) Musil, F.; Grisafi, A.; Bartók, A. P.; Ortner, C.; Csányi, G.; Ceriotti, M. Physics-Inspired Structural Representations for Molecules and Materials. *Chem. Rev.* **2021**, *121*, 9759–9815.

(5) Behler, J. Atom-Centered Symmetry Functions for Constructing High-Dimensional Neural Network Potentials. *J. Chem. Phys.* **2011**, *134*, 074106.

(6) Bartók, A. P.; Kondor, R.; Csányi, G. On Representing Chemical Environments. *Phys. Rev. B* **2013**, *87*, 184115.

(7) Thompson, A. P.; Swiler, L. P.; Trott, C. R.; Foiles, S. M.; Tucker, G. J. Spectral Neighbor Analysis Method for Automated Generation of Quantum-Accurate Interatomic Potentials. *J. Comput. Phys.* **2015**, *285*, 316–330.

(8) Shapeev, A. V. Moment Tensor Potentials: A Class of Systematically Improvable Interatomic Potentials. *Multiscale Model. Simul.* **2016**, *14*, 1153–1173.

(9) Drautz, R. Atomic Cluster Expansion for Accurate and Transferable Interatomic Potentials. *Phys. Rev. B* **2019**, *99*, 014104.

(10) Dusson, G.; Bachmayr, M.; Csányi, G.; Drautz, R.; Etter, S.; van der Oord, C.; Ortner, C. Atomic Cluster Expansion: Completeness, Efficiency and Stability. *J. Comput. Phys.* **2022**, *454*, 110946.

(11) Lysogorskiy, Y.; others Performant Implementation of the Atomic Cluster Expansion (PACE) and Application to Copper and Silicon. *npj Comput. Mater.* **2021**, *7*, 97.

(12) Schütt, K. T.; Sauceda, H. E.; Kindermans, P.-J.; Tkatchenko, A.; Müller, K.-R. SchNet – A Deep Learning Architecture for Molecules and Materials. *J. Chem. Phys.* **2018**, *148*, 241722.

(13) Unke, O. T.; Meuwly, M. PhysNet: A Neural Network for Predicting Energies, Forces, Dipole Moments, and Partial Charges. *J. Chem. Theory Comput.* **2019**, *15*, 3678–3693.

(14) Gasteiger, J.; Groß, J.; Günnemann, S. Directional Message Passing for Molecular Graphs. International Conference on Learning Representations. 2020; arXiv:2003.03123.

(15) Thomas, N.; Smidt, T.; Kearnes, S.; Yang, L.; Li, L.; Kohlhoff, K.; Riley, P. Tensor Field Networks: Rotation- and Translation-Equivariant Neural Networks for 3D Point Clouds. 2018; arXiv:1802.08219.

(16) Schütt, K. T.; Unke, O. T.; Gastegger, M. Equivariant Message Passing for the Prediction of Tensorial Properties and Molecular Spectra. Proceedings of the 38th International Conference on Machine Learning. 2021; pp 9377–9388.

(17) Batzner, S.; Musaelian, A.; Sun, L.; Geiger, M.; Mailoa, J. P.; Kornbluth, M.; Molinari, N.; Smidt, T. E.; Kozinsky, B. E(3)-Equivariant Graph Neural Networks for Data-Efficient and Accurate Interatomic Potentials. *Nat. Commun.* **2022**, *13*, 2453.

(18) Musaelian, A.; Batzner, S.; Johansson, A.; Sun, L.; Owen, C. J.; Kornbluth, M.; Kozinsky, B. Learning Local Equivariant Representations for Large-Scale Atomistic Dynamics. *Nat. Commun.* **2023**, *14*, 579.

(19) Liao, Y.-L.; Smidt, T. Equiformer: Equivariant Graph Attention Transformer for 3D Atomistic Graphs. International Conference on Learning Representations. 2023; arXiv:2206.11990.

(20) Frank, J. T.; Unke, O. T.; Müller, K.-R. So3krates: Equivariant Attention for Interactions on Arbitrary Length-Scales in Molecular Systems. Advances in Neural Information Processing Systems. 2022; arXiv:2205.14276.

(21) Wang, Y.; others Enhancing Geometric Representations for Molecules with Equivariant Vector-Scalar Interactive Message Passing. *Nat. Commun.* **2024**, *15*, 313.

(22) Wang, Z.; others Efficiently Incorporating Quintuple Interactions into Geometric Deep Learning Force Fields. Advances in Neural Information Processing Systems. 2023.

(23) Aykent, S.; Xia, T. GotenNet: Rethinking Efficient 3D Equivariant Graph Neural Networks. International Conference on Learning Representations. 2025.

(24) Batatia, I.; Kovács, D. P.; Simm, G. N. C.; Ortner, C.; Csányi, G. MACE: Higher Order Equivariant Message Passing Neural Networks for Fast and Accurate Force Fields. Advances in Neural Information Processing Systems. 2022; arXiv:2206.07697.

(25) Bochkarev, A.; Lysogorskiy, Y.; Drautz, R. Graph Atomic Cluster Expansion for Semilocal Interactions beyond Equivariant Message Passing. *Phys. Rev. X* **2024**, *14*, 021036.

(26) Batatia, I.; others The Design Space of E(3)-Equivariant Atom-Centered Interatomic Potentials. 2022; arXiv:2205.06643.

(27) Chmiela, S.; Tkatchenko, A.; Sauceda, H. E.; Poltavsky, I.; Schütt, K. T.; Müller, K.-R. Machine Learning of Accurate Energy-Conserving Molecular Force Fields. *Sci. Adv.* **2017**, *3*, e1603015.

(28) Christensen, A. S.; von Lilienfeld, O. A. On the Role of Gradients for Machine Learning of Molecular Energies and Forces. *Mach. Learn.: Sci. Technol.* **2020**, *1*, 045018.

(29) Passaro, S.; Zitnick, C. L. Reducing SO(3) Convolutions to SO(2) for Efficient Equivariant GNNs. 2023; arXiv:2302.03655.

(30) Luo, S.; Chen, T.; Krishnapriyan, A. S. Enabling Efficient Equivariant Operations in the Fourier Basis via Gaunt Tensor Products. 2024; arXiv:2401.10216.

(31) Simeon, G.; De Fabritiis, G. TensorNet: Cartesian Tensor Representations for Efficient Learning of Molecular Potentials. Advances in Neural Information Processing Systems. 2023; arXiv:2306.06482.

(32) Cheng, B. Cartesian Atomic Cluster Expansion for Machine Learning Interatomic Potentials. *npj Comput. Mater.* **2024**, arXiv:2402.07472.

(33) Zaverkin, V.; others Higher-Rank Irreducible Cartesian Tensors for Equivariant Message Passing. 2024; NeurIPS 2024; arXiv:2405.14253.

(34) Zuo, Y.; Chen, C.; Li, X.; Deng, Z.; Chen, Y.; Behler, J.; Csányi, G.; Shapeev, A. V.; Thompson, A. P.; Wood, M. A.; Ong, S. P. Performance and Cost Assessment of Machine Learning Interatomic Potentials. *J. Phys. Chem. A* **2020**, *124*, 731–745.

(35) Grisafi, A.; Ceriotti, M. Incorporating Long-Range Physics in Atomic-Scale Machine Learning. *J. Chem. Phys.* **2019**, *151*, 204105.

(36) Ko, T. W.; Finkler, J. A.; Goedecker, S.; Behler, J. A Fourth-Generation High-Dimensional Neural Network Potential with Accurate Electrostatics Including Non-Local Charge Transfer. *Nat. Commun.* **2021**, *12*, 398.

(37) Unke, O. T.; Chmiela, S.; Gastegger, M.; Schütt, K. T.; Sauceda, H. E.; Müller, K.-R. SpookyNet: Learning Force Fields with Electronic Degrees of Freedom and Nonlocal Effects. *Nat. Commun.* **2021**, *12*, 7273.

(38) Kosmala, A.; Gasteiger, J.; Gao, N.; Günnemann, S. Ewald-Based Long-Range Message Passing for Molecular Graphs. Proceedings of the 40th International Conference on Machine Learning. 2023; arXiv:2303.04791.

(39) Cheng, B. Latent Ewald Summation for Machine Learning of Long-Range Interactions. 2024; arXiv:2408.15165; npj Comput. Mater. 2025.

(40) Lakshminarayanan, B.; Pritzel, A.; Blundell, C. Simple and Scalable Predictive Uncertainty Estimation Using Deep Ensembles. Advances in Neural Information Processing Systems. 2017.

(41) Hinton, G.; Vinyals, O.; Dean, J. Distilling the Knowledge in a Neural Network. 2015; arXiv:1503.02531.

(42) Gardner, J. L. A.; others 2025; Teacher-to-student distillation of atomistic models; arXiv:2506.10956.

(43) GRACE developers 2025; Distillation into graph atomic cluster expansion students; arXiv:2508.17936.

(44) Essmann, U.; Perera, L.; Berkowitz, M. L.; Darden, T.; Lee, H.; Pedersen, L. G. A Smooth Particle Mesh Ewald Method. *J. Chem. Phys.* **1995**, *103*, 8577–8593.

(45) Furlanello, T.; Lipton, Z. C.; Tschannen, M.; Itti, L.; Anandkumar, A. Born-Again Neural Networks. Proceedings of the 35th International Conference on Machine Learning. 2018; arXiv:1805.04770.

(46) Elfwing, S.; Uchibe, E.; Doya, K. Sigmoid-Weighted Linear Units for Neural Network Function Approximation in Reinforcement Learning. *Neural Netw.* **2018**, *107*, 3–11.

(47) Darby, J. P.; others Tensor-Reduced Atomic Density Representations. 2022; arXiv:2210.01705.

(48) Kolda, T. G.; Bader, B. W. Tensor Decompositions and Applications. *SIAM Rev.* **2009**, *51*, 455–500.

(49) Pozdnyakov, S. N.; Willatt, M. J.; Bartók, A. P.; Ortner, C.; Csányi, G.; Ceriotti, M. Incompleteness of Atomic Structure Representations. *Phys. Rev. Lett.* **2020**, *125*, 166001.

(50) Ba, J. L.; Kiros, J. R.; Hinton, G. E. Layer Normalization. 2016; arXiv:1607.06450.

(51) Rappé, A. K.; Goddard, W. A. Charge Equilibration for Molecular Dynamics Simulations. *J. Phys. Chem.* **1991**, *95*, 3358–3363.

(52) Ewald, P. P. Die Berechnung optischer und elektrostatischer Gitterpotentiale. *Ann. Phys.* **1921**, *369*, 253–287.

(53) Darden, T.; York, D.; Pedersen, L. Particle Mesh Ewald: An N·log(N) Method for Ewald Sums in Large Systems. *J. Chem. Phys.* **1993**, *98*, 10089–10092.

(54) Tkatchenko, A.; Scheffler, M. Accurate Molecular Van Der Waals Interactions from Ground-State Electron Density and Free-Atom Reference Data. *Phys. Rev. Lett.* **2009**, *102*, 073005.

(55) Grimme, S.; Ehrlich, S.; Goerigk, L. Effect of the Damping Function in Dispersion Corrected Density Functional Theory. *J. Comput. Chem.* **2011**, *32*, 1456–1465.

(56) Grimme, S.; Antony, J.; Ehrlich, S.; Krieg, H. A Consistent and Accurate Ab Initio Parametrization of Density Functional Dispersion Correction (DFT-D) for the 94 Elements H–Pu. *J. Chem. Phys.* **2010**, *132*, 154104.

(57) Ziegler, J. F.; Biersack, J. P.; Littmark, U. *The Stopping and Range of Ions in Solids*; Pergamon: New York, 1985.

(58) Kingma, D. P.; Ba, J. Adam: A Method for Stochastic Optimization. 2015; ICLR 2015; arXiv:1412.6980.

(59) Reddi, S. J.; Kale, S.; Kumar, S. On the Convergence of Adam and Beyond. International Conference on Learning Representations. 2018.

(60) Loshchilov, I.; Hutter, F. Decoupled Weight Decay Regularization. 2019; ICLR 2019; arXiv:1711.05101.

(61) Smith, L. N.; Topin, N. Super-Convergence: Very Fast Training of Neural Networks Using Large Learning Rates. 2017; arXiv:1708.07120.

(62) Geiger, M.; Smidt, T. e3nn: Euclidean Neural Networks. 2022; arXiv:2207.09453.

(63) Yang, G.; others Tensor Programs V: Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer. 2022; arXiv:2203.03466.

(64) Leimkuhler, B.; Matthews, C. Rational Construction of Stochastic Numerical Methods for Molecular Sampling. *Appl. Math. Res. Express* **2013**, *2013*, 34–56.

(65) Paszke, A.; others PyTorch: An Imperative Style, High-Performance Deep Learning Library. Advances in Neural Information Processing Systems. 2019.

(66) Okuta, R.; Unno, Y.; Nishino, D.; Hido, S.; Loomis, C. CuPy: A NumPy-Compatible Library for NVIDIA GPU Calculations. Proceedings of Workshop on Machine Learning Systems (LearningSys) at NIPS. 2017.

(67) Larsen, A. H.; others The Atomic Simulation Environment—A Python Library for Working with Atoms. *J. Phys.: Condens. Matter* **2017**, *29*, 273002.

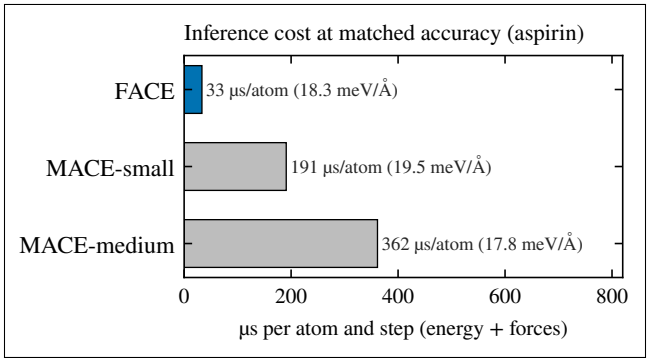
(68) Fu, X.; Wu, Z.; Wang, W.; Xie, T.; Keten, S.; Gómez-Bombarelli, R.; Jaakkola, T. Forces Are Not Enough: Benchmark and Critical Evaluation for Machine Learning Force Fields with Molecular Simulations. 2023; Trans. Mach. Learn. Res.; arXiv:2210.07237.

(69) Bigi, F.; Langer, M. F.; Ceriotti, M. The Dark Side of the Forces: Assessing Non-Conservative Force Models for Atomistic Machine Learning. 2024; ICML 2025; arXiv:2412.11569.

(70) Kovács, D. P.; others Linear Atomic Cluster Expansion Force Fields for Organic Molecules: Beyond RMSE. *J. Chem. Theory Comput.* **2021**, *17*, 7696–7711.

(71) Bigi, F.; others A Prediction Rigidity Formalism for Low-Cost Uncertainties in Trained Neural Networks. 2024; arXiv:2403.02251.

TOC Graphic


Inference cost at matched accuracy (aspirin)
FACE
33 µs/atom (18.3 meV/Å)
MACE-small
191 µs/atom (19.5 meV/Å)
MACE-medium
362 µs/atom (17.8 meV/Å)
0
200
400
600
800
µs per atom and step (energy + forces)